S226 PRT661 DATA SCIENCE PRACTICE

Assessment 4 Group Project Final Report
Danala Group 8 Theme 2

Australian NEM Price & Demand Analysis and Forecasting System

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Jira project board:

https://shrestharoshan931-1785816864381.atlassian.net/jira/software/projects/KAN/list

October 2026

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Section 1 Assessment 1 and 2

S226 PRT661 DATA SCIENCE PRACTICE

Assessment 2 Group Project Progress Report
Danala Group 8 Theme 2

Electricity Demand and Price Forecasting for the Australian National Electricity Market

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September 2026

Section 1 – Assessment 1 Report (Original Proposal)

S226 PRT661 DATA SCIENCE PRACTICE

Assessment 1 Group Project Proposal
Danala Group 8 Theme 2

Electricity Demand and Price Forecasting for the Australian National Electricity Market

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1. Project Overview and Objectives

This project aims to develop an end-to-end predictive analytics system that will offer forecasts of regional electricity demand and wholesale electricity prices within the 5 National Electricity Market (NEM) regions (QLD1, NSW1, VIC1, SA1 and TAS1). The system will be based on historical and the latest data released by the Australian Energy Market Operator (AEMO) via NEMWEB. The integration of demand, price and generation data will be connected with the common time-series data of rooftop PV and interconnector-flow.

The above mentioned pipeline is proposed to include data acquisition, ETL processing, storage, feature engineering, model development, evaluation, forecasting and visualisation. Automated workflow orchestration will be done using Apache Airflow, local relational storage will be enabled by SQL Server, the development of the LSTM/GRU model will be done using TensorFlow and the final interactive visualization will be done using Tableau Public.

The project has 4 key objectives:

- Make short-term multi-horizon predictions of regional electricity demand and Regional Reference Price (RRP).
- Include more explanatory variables, such as generation mix, rooftop PV and interconnector flows.
- Set up an automated Airflow pipeline for data ingestion, transformation, storage and forecast generation.
- Create Tableau Public dashboards of actual vs. forecast values, regional comparisons, and model performance.

2. Theme Justification

The project fits well within the scope of Theme 2 – Predictive Analytics and Forecasting as its main goal is to predict the future electricity demand and wholesale prices, given past observations and explanatory variables. The proposed workflow consists of data acquisition, data preparation, time-series feature engineering, model development, evaluation and visualization, which is directly related to the requirements of the chosen theme.

The LSTM and GRU models will be explored since they are specifically created for sequential data and learn temporal dependencies (Hochreiter & Schmidhuber, 1997; Cho et al., 2014). Feature construction will be based on established forecasting principles, as will the chronological validation and evaluation of the forecasts (Hyndman & Athanasopoulos, 2021).

3. Problem Statement

The utility of short term forecasts for monitoring and decision making in the NEM is important. While AEMO offers well established market forecasts, this project will explore the possibility of making improvements to this forecasting through the use of the integration of demand, price, generation, rooftop PV and interconnector characteristics with machine learning models.

The project will create an independent forecasting system based on the data from NEMWEB for the five NEM regions. Observations will be cleaned and merged, and then converted to time-series features like lagged demand and price, rolling statistics and calendar variables. LSTM/GRU models will produce multi-horizon forecasts for the demand and RRP.

The project will evaluate the feasibility and predictive performance of this feature-based automated forecasting approach against relevant forecasting baselines.

4. Initial Architecture Diagram

The system will follow a layered architecture, orchestrated end-to-end by Apache Airflow:

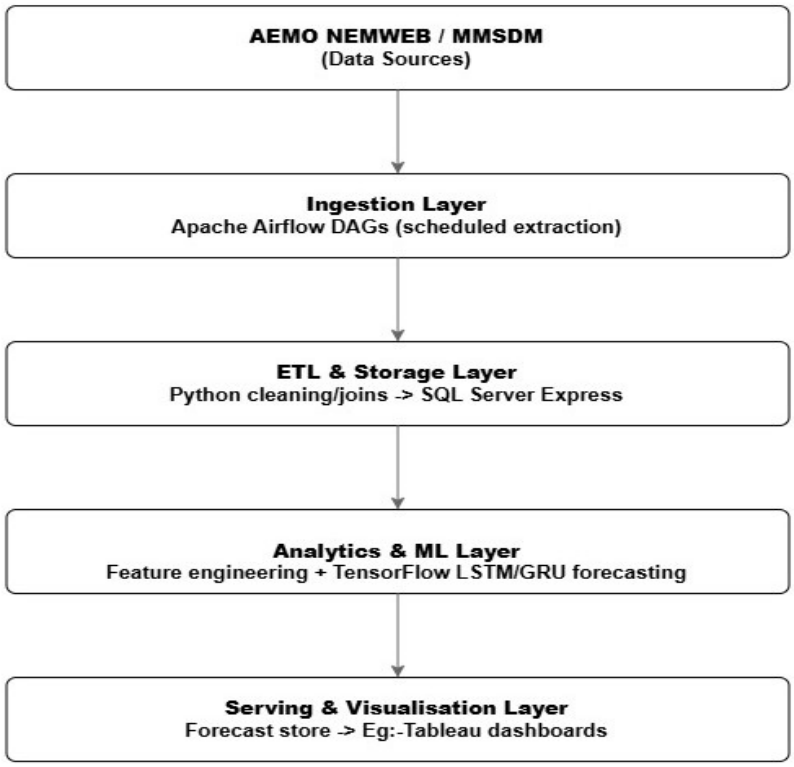

Figure 1 Layered System Architecture for the NEM Forecasting Pipeline

4.1 Workflow Plan

The proposed system is based on a layer structure, which comprises data acquisition, processing, storage, forecasting and visualisation. The workflow will be orchestrated by Apache Airflow and data processed and models developed using Python. The forecasting results will be presented using Tableau Public and SQL Server Express will be used to provide relational storage.

The workflow will now go through the following steps:

- **Data Acquisition:** The data will be acquired using the open source Python package, NEMOSIS (UNSW Collaboration on Energy and Environmental Markets, n.d.), which is a tool for downloading and compiling market data tables from AEMO that are publicly available as AEMWEB. This allows regional demand, price, rooftop PV and interconnector datasets to be automatically retrieved for the desired historical and current time periods.
- **Data Cleaning:** Cleaning the downloaded data by eliminating duplicate data, validating timestamps, checking for gaps.
- **Feature engineering** variables, which includes calendar variables, lagged demand/price values, rolling statistics etc. This will capture temporal patterns, as well as minimize the chance of data leakage (Hyndman & Athanasopoulos, 2021).
- **Data Storage:** The cleaned and feature-enhanced data will be saved to SQL Server to be used for model training, analysis and generastion of forecasts.
- Chronological historical sequences will be fed to LSTM and GRU models for learning temporal dependencies (Hochreiter & Schmidhuber, 1997; Cho et al., 2014).

- **Evaluation** of forecasts will be based on MAE, RMSE and MAPE on held-out chronological data to evaluate it. Baseline forecasts will also be used to assess whether or not the models proposed are meaningful predictive improvement (Hyndman & Koehler, 2006).
- The developed models will provide forecasts of demand and price.
- **Automation:** Apache Airflow handles task dependency, scheduling, retries and logging, allowing the pipeline to be automatically run when new data is available.

5. Project Governance and Milestones

Communication and channel: weekly team meetings, and channel for the team in Microsoft teams for the day-to-day communication and file sharing.

Repository and tracking: a shared GitHub repository, and a Jira Kanban board for sprint-level task tracking.

Conflict Resolution: Major decisions on technical and scope are agreed by consensus, Project Lead has overall time line and documentation. First technical differences are discussed within the workstream and, if not resolved, are moved to the Project Lead for decision making, while involving the whole team in major changes to the architecture or scope.

All of the code, diagrams and reports are stored in a version-controlled repository on GitHub and progress is tracked every week, based on the milestones below.

Table 1 High-Level Milestones

Phase	Milestone & Lead
Phase 1 – Foundation (Wks 1–2)	ALL Members are involved. Nitesh is responsible for setting up repositories and repository architecture, Roshan for data-source confirmation, Dilliram and Dipesh for data exploration and set-up of data-boards.
Phase 2 – Data Pipeline (Wks 3–4)	All members involved. Dipesh is responsible for Airflow ETL and SQL Server loading, Dilliram for pipeline design, Roshan and Nitesh for data cleaning and quality checks.
Phase 3 – Modelling (Wks 5–7)	All members involved. Roshan leads TensorFlow LSTM/GRU development; Nitesh supports evaluation metrics; Dipesh and Dilliram assist with feature engineering and review.
Phase 4 – Forecasting & Visualisation (Wks 8–9)	All members involved. Dilliram leads forecasting and the Tableau dashboard; Roshan supports residual analysis; Dipesh and Dilliram contribute to validation and documentation.
Phase 5 – Integration & Finalisation (Wks 10–11)	Integration & Finalisation (Wks 10–11), All members involved. Full responsibility for pipeline integration, as well as joint testing, final pipeline documentation and presentation preparation.

5.1 Repository Structure and Coding Standards

The codebase will be designed in an object-oriented, modular fashion in order to allow the development, testing and re-use of components for ingestion, transformation, modelling and serving. Coding standards: Planned repository layout:

```
nem-forecasting/x
 |-- src/
 |   |-- ingestion/      (NemwebClient, DagTasks)
 |   |-- etl/            (Transformer, FeatureBuilder)
 |   |-- models/         (BaseForecaster, LstmForecaster)
 |   |-- storage/        (SqlServerRepository)
 |   +-- visualisation/  (TableauExporter)
 |-- dags/               (Airflow DAG definitions)
 |-- tests/              (pytest unit/integration tests)
```

```
|-- docs/            (architecture.drawio, workflow.drawio)
+-- README.md
```

6. Task Allocation and Team Structure

Table 2 Team Roles and Responsibilities

Member	Role	Key Responsibilities
Nitesh	Project Lead & Architecture	Overall coordination, architecture diagrams, proposal writing, GitHub repository management, risk and ethics oversight.
Dipesh	Data Engineer	Airflow DAGs, NEMWEB data-download scripts, ETL pipelines, multi-table joins, SQL Server loading.
Roshan	ML Engineer (TensorFlow)	Sequence feature engineering, TensorFlow LSTM model development, tuning and evaluation.
Dilliram	Forecasting, Validation & Dashboard	Forecast generation, with AEMO predispatch, residual analysis, Tableau Public dashboards, documentation.

7. Initial Risk Analysis

Table 3 Initial Risk Analysis

Risk	Likelihood / Impact	Mitigation
Data download failures / NEMWEB volume	Medium / Medium	Incremental daily downloads, Airflow retry logic, local caching.
Missing or incomplete data intervals	Medium / Medium	Handling missing data and automated data quality checks in the Transform stage.
Model performance degradation over time	Medium / High	Retrain and continually check performance against recent actuals.
Computational limits on student machines	High / Medium	Focus on recent data; efficient sequence windows and batch sizes.
Team coordination and knowledge silos	Low / Medium	having clear team roles, shared GitHub repository, weekly progress reviews.

8. Ethics, Privacy and Security Considerations

AEMO releases all data publicly, and does not include personal, commercially sensitive or confidential information.

The project will:

- Correctly cite AEMO and any other resources as data source in all reports/visualisations
- Store securely data and models in private GitHub repository/free tier/ local environments only
- clearly describe the model's limitations and the uncertainty of predictions, to prevent over-optimism about predictions
- Follow academic integrity and University data handling policies.

9. Gantt Chart

Phase	Duration	Weeks										
		Aug 2026				Sep 2026				Oct 2026		
		Wk 1	Wk 2	Wk 3	Wk 4	Wk 5	Wk 6	Wk 7	Wk 8	Wk 9	Wk 10	Wk 11
Phase 1 – Foundation	Wks 1–2	■	■									
Phase 2 – Data Pipeline	Wks 3–4			■	■							
Phase 3 – Modelling	Wks 5–7					■	■	■				
Phase 4 – Forecasting & Visualisation	Wks 8–9								■	■		
Phase 5 – Integration & Finalisation	Wks 10–11										■	■

Figure 2 Project Gantt Chart

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Appendices

Github repo link: https://github.com/niteshS394811/S226-PRT661-DATA-SCIENCE-PRACTICE

Draw.io file link: https://drive.google.com/file/d/1IfyWJmD-7Q-5yMSFf5EfJbY1dxM3Vrfw/view

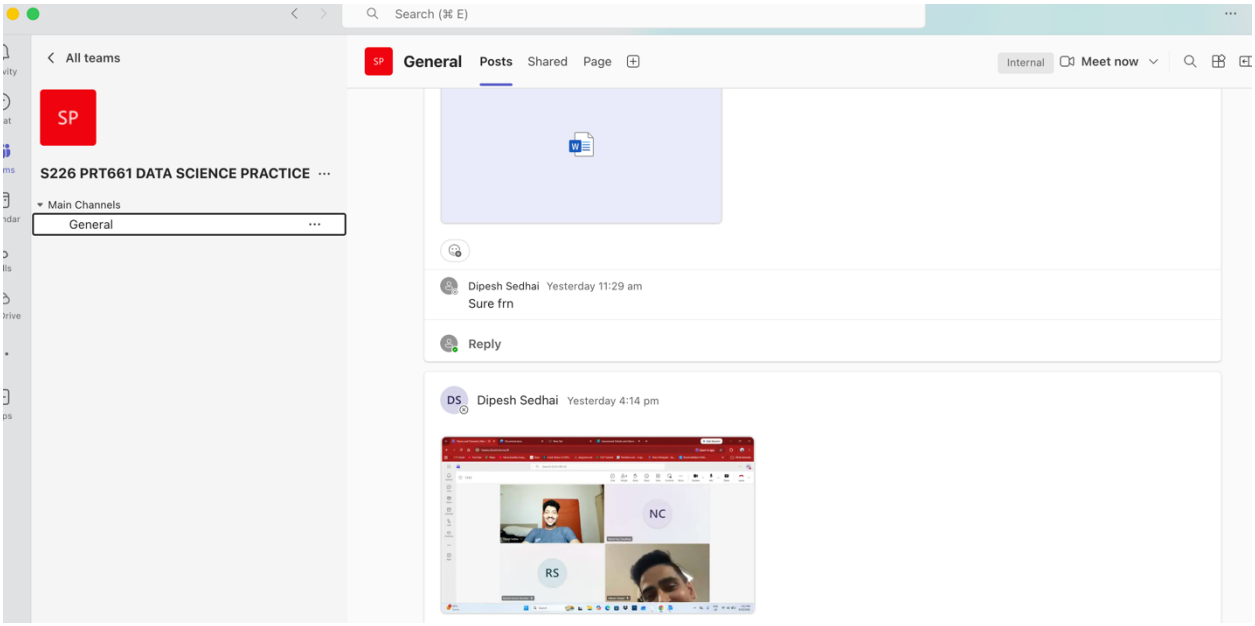


Figure 3 Teams communication channel

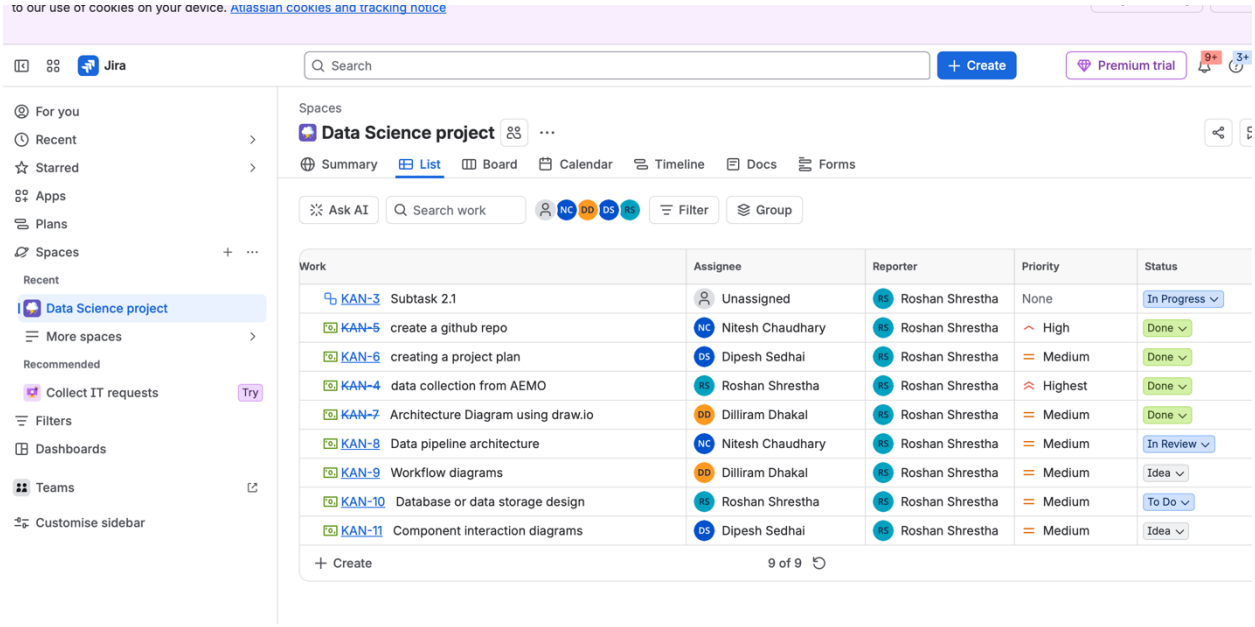


Figure 4 Jira for project governance

Section 2 – Project Update Summary

Changes Since Previous Assessment

Area	Previous State (A1)	Current State	Justification
Storage	SQL Server Express	PostgreSQL 17 in Docker; staging→dwh→datamart	Portable across members; production-style warehouse
Orchestration	Airflow planned	Airflow in Docker + CLI scripts	Reproducible; CLI fallback when laptop is offline
Visualisation	Tableau Public	2 Streamlit apps (:8501 ops, :8502 models)	Direct Postgres; no external licensing
Modelling	TensorFlow LSTM/GRU	Seasonal Naive, Ridge, HistGradientBoosting	Tractable on student hardware; temporal signal via lags
Pipelines	Single pipeline described	History (full+train), daily (upsert), weekly (retrain)	Clear separation; reproducible catch-up loads

Not delivered versus A1 (justified): Tableau; SQL Server; LSTM/GRU as primary models; rooftop PV as a core feature. Scope was narrowed to a working end-to-end warehouse and tabular multi-model comparison but can be extended in further assessment.

Contribution Summary

Team Member	Contribution Since Assessment 1
Dilliram Dhakal (S396633)	Extraction date-range fixes; price/demand cleaning; forecast and dashboard validation
Dipesh Sedhai (S395457)	Postgres incremental loading; merge conflict resolution; ETL support
Nitesh Raj Chaudhary (S394811)	Extraction updates; Docker/Postgres schemas; scaffold, dependencies, repo/report coordination
Roshan Kumar Shrestha (S395498)	Dockerfile configuration; datamart load; branch integration; modelling support

Evidence Locations

Evidence	Location
Source code	GitHub – /src, /scripts, dashboard_daily.py, dashboard_weekly.py
Docker / orchestration	docker-compose.yml, Dockerfile, Dockerfile.airflow, /docker/postgres
Commit history	https://github.com/niteshS394811/S226-PRT661-DATA-SCIENCE-PRACTICE/commits/main
Task tracking	Jira – Data Science Project board (KAN)
Reports	Repo root assessment documents

Section 3 – Technical Development (Since Assessment 1)

3.1 Updated System Architecture

Assessment 1 proposed: NEMWEB → Airflow → SQL Server → TensorFlow LSTM/GRU → Tableau. The implemented system keeps the layered philosophy with a more tractable stack:

NEMWEB (AEMO) → NEMOSIS extract → clean → staging.stg_price_demand → dwh.panel / dwh.features → datamart (actuals, metrics, forecasts) → Streamlit dashboards. History, daily and weekly pipelines orchestrate these steps; Apache Airflow can trigger the same logic as CLI scripts under /scripts.

All services are containerized via Docker Compose: a custom PostgreSQL 17 image, three Airflow services (scheduler, API server, DAG processor), and two Streamlit dashboard containers. This makes the environment reproducible across all four team members' machines – a specific weakness of the original SQL Server Express plan, which was harder to run consistently outside Windows.

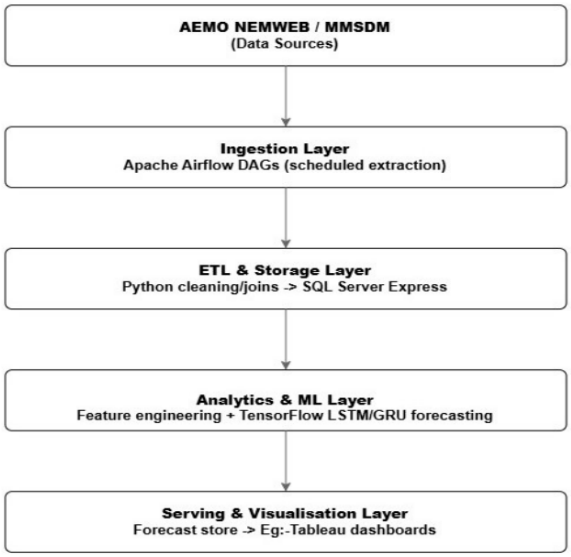


Figure 1: original proposed architecture

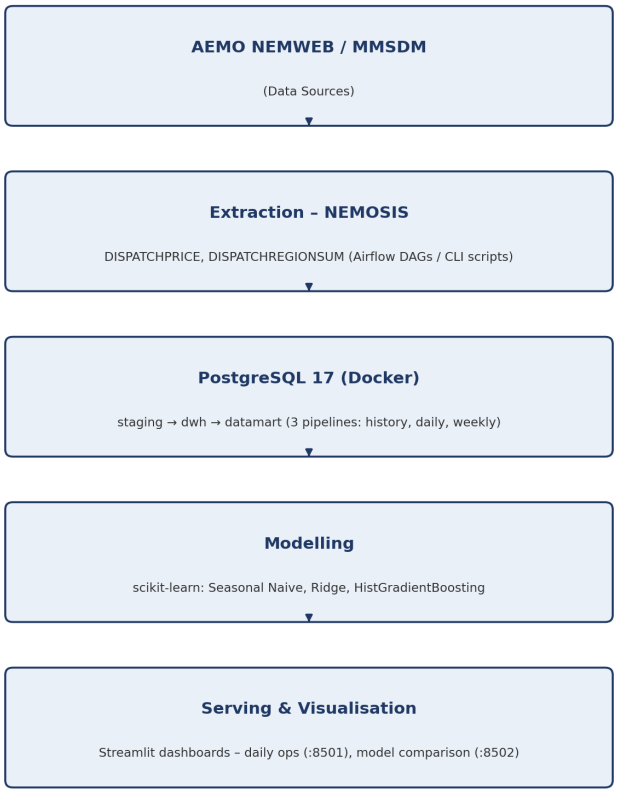


Figure 2 (Assessment 2): updated, implemented architecture

Services are containerised via Docker Compose: custom PostgreSQL 17, Airflow (scheduler, API, DAG processor), and two Streamlit containers — reproducible across team machines.

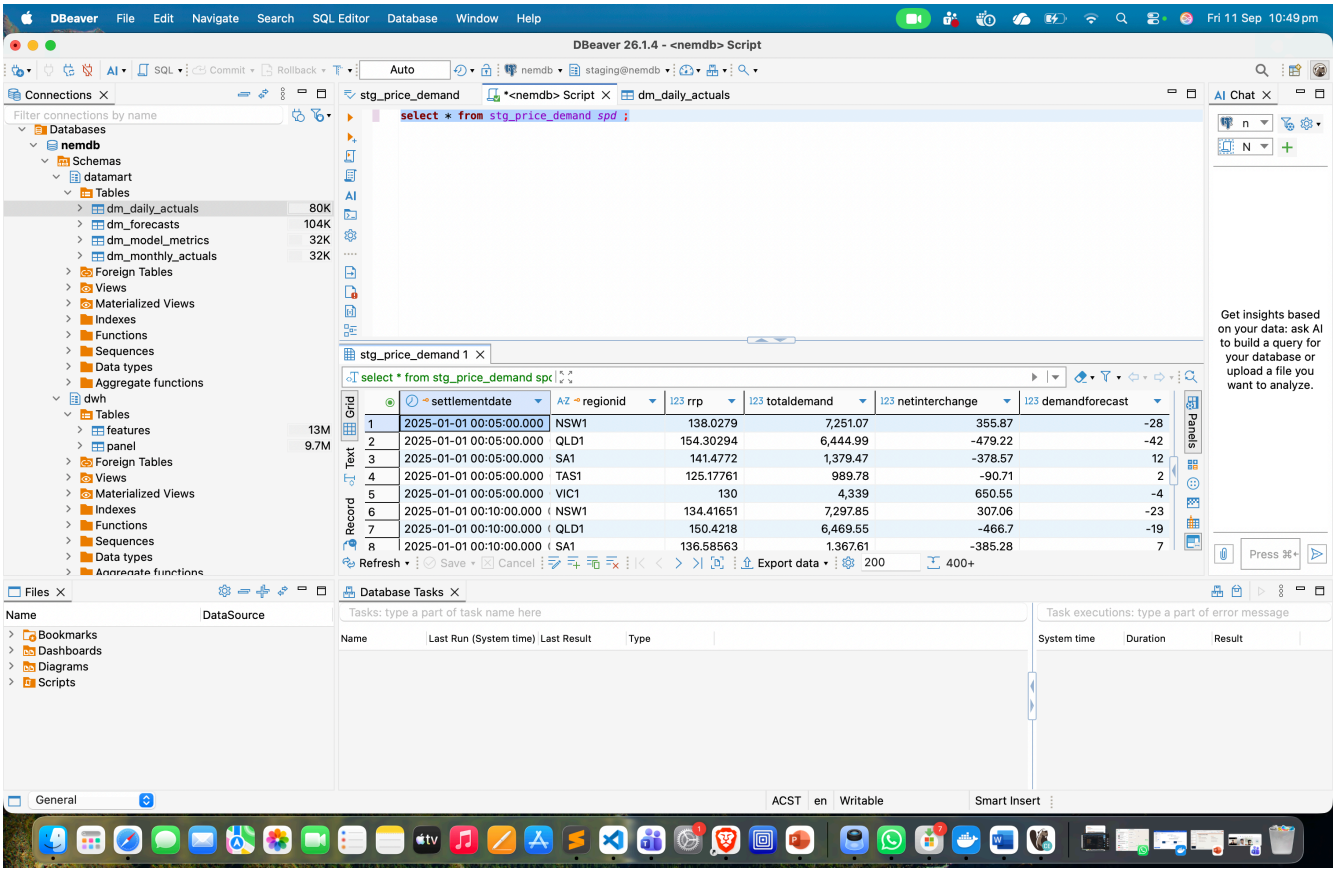
3.2 Data Acquisition and Storage

Data is sourced from AEMO via NEMWEB using NEMOSIS dynamic_data_compiler for DISPATCHPRICE and DISPATCHREGIONSUM. Rows with INTERVENTION = 1 are filtered out. Regions: NSW1, QLD1, VIC1, SA1, TAS1 at 5-minute grain.

Columns used:

- RRP - Regional Reference Price ($/MWh).
- TOTALDEMAND - operational demand (MW) - primary demand target.
- NETINTERCHANGE - net inter-regional power flow (MW); positive ≈ import.
- DEMANDFORECAST - AEMO's 5-minute demand adjustment (offset), not an absolute MW demand forecast. Official MMS documentation labels this field as a "5 minute forecast adjust". Values are typically tens of MW (e.g. −28), not thousands. Therefore aemo_demand_error = totaldemand − demandforecast must not be interpreted as absolute AEMO demand-forecast error. Absolute operational demand forecasts would require other AEMO tables (e.g. pre-dispatch / operational demand forecast packages), noted as future work.

Schema	Tables	Role
staging	stg_price_demand	Cleaned extract; incremental upsert
dwh	panel, features	Calendar fields, lags, rolling statistics
datamart	dm_daily_actuals, dm_monthly_actuals, dm_forecasts, dm_model_metrics (+ hourly forecast tables where enabled)	Reporting and model outputs



SELECT settlementdate::date, COUNT(*) FROM staging.stg_price_demand GROUP BY 1 ORDER BY 1;

3.3 Data Engineering: Three Pipelines

Pipeline	Behaviour	Trains model?
History	Full refresh for a date range; rebuilds all layers	Yes (5-min models; hourly 1d/1w where enabled)
Daily	MAX(settlementdate)+1 day; upsert that day only	No full retrain; can refresh hourly forecasts
Weekly	Retrain on full features; write forecasts	Yes

Daily upsert deletes the full calendar-day window (midnight bounds) before insert to avoid primary-key collisions and keep loads idempotent.

3.4 Data Cleaning and Preprocessing

Cleaning and preprocessing are performed at two stages: when staging is performed as a transform in the DWH, and just before model training. The categories are each discussed in detail below, in terms of the rubrics.

- Duplicates: timestestamps of dispatched rows are deduplicated when extracted and the full target day is deleted prior to re-insertion in the daily pipeline, which means that a load will never result in duplicate rows on the same target day.
- There are no rows loaded in the warehouse that have an intervention value of 1, as these are considered incorrect / invalid records and will not be market outcomes but manual dispatch interventions by the AEMO.
- Early history lag features (e.g. demand_lag_288) will be undefined for the first day of each load. The rows are dropped using dropna before fitting Ridge, which can't accept NaNs; fillna(0) for demand_roll_std_12 where a short window results in an undefined rolling standard deviation.
- Data types: settlementdate is explicitly parsed to a correct timestamp type on extraction (not as a string) which is essential for date-range math in the daily incremental pipeline and for time ordering of data for a hold-out/CV split.

- No outlier removal for RRP as true price spikes in the NEM are a meaningful signal the models are trying to learn – removing them would introduce bias to the models away from the behaviour they need.
- In the Encoding section regionid is used as is, as a partition key (one model trained per region) and not as a one-hot encoding because a model trained per region helps avoid confusing structurally different regions (e.g. TAS1 vs NSW1 volatility).
- Scaling – in this iteration features are not rescaled (sensible results without scaling up/down were checked for HistGradientBoosting; standardising inputs before Ridge mentioned in Section 3.11 as concrete next step).
- Bug fix – UniqueViolation on incremental load: This bug was triggered by the delete window not being aligned with calendar day boundaries; it has been fixed by normalising all delete/upsert windows to midnight-to-midnight before insert.
- Bug fix – driver mismatch: fixed the SQLAlchemy/psycopg connection string (postgresql+psycopg) to be standardized amongst all services following a dialect mismatch with psycopg2 vs psycopg3.

3.5 Feature Engineering

Feature engineering follows the Assessment 1 plan (lagged demand/price, rolling statistics, calendar variables) and adds an interchange-based feature:

- Calendar: hour, day_of_week, is_weekend.
- Demand lags: 1 step (~5 min), 12 steps (~1 hour), 288 steps (~1 day).
- Price lags: 1 and 12 steps.
- Rolling statistics: demand mean and standard deviation over a 12-interval window.
- Net interchange and its 1-step lag – added beyond the original plan to capture inter-regional import/export pressure.

3.6 Analytics, Visualisation, and Model Experimentation

There are three models trained per region (per target): A baseline model, a Seasonal Naive model, Ridge regression (L2-regularised linear model on engineered features), and HistGradientBoosting (HistGB – a non-linear tree ensemble, which is the main model that is used for live forecasts). This is instead of the Assessment 1 plan to utilize TensorFlow sequence models, specifically LSTM/GRU, which were found to be impractical within the time frame of the semester and the computational power of students' laptops, as stated in the Assessment 1 risk register.

Evaluation is done through a time ordered hold-out split plus walk-forward cross-validation, and the metrics will be stored in datamart.dm_model_metrics for the dashboards, including MAE, RMSE, cv_mae, and cv_rmse. The best hold-out model per region for the demand target for the latest training run is as follows: (**Note: Result may vary according to date range load.)**

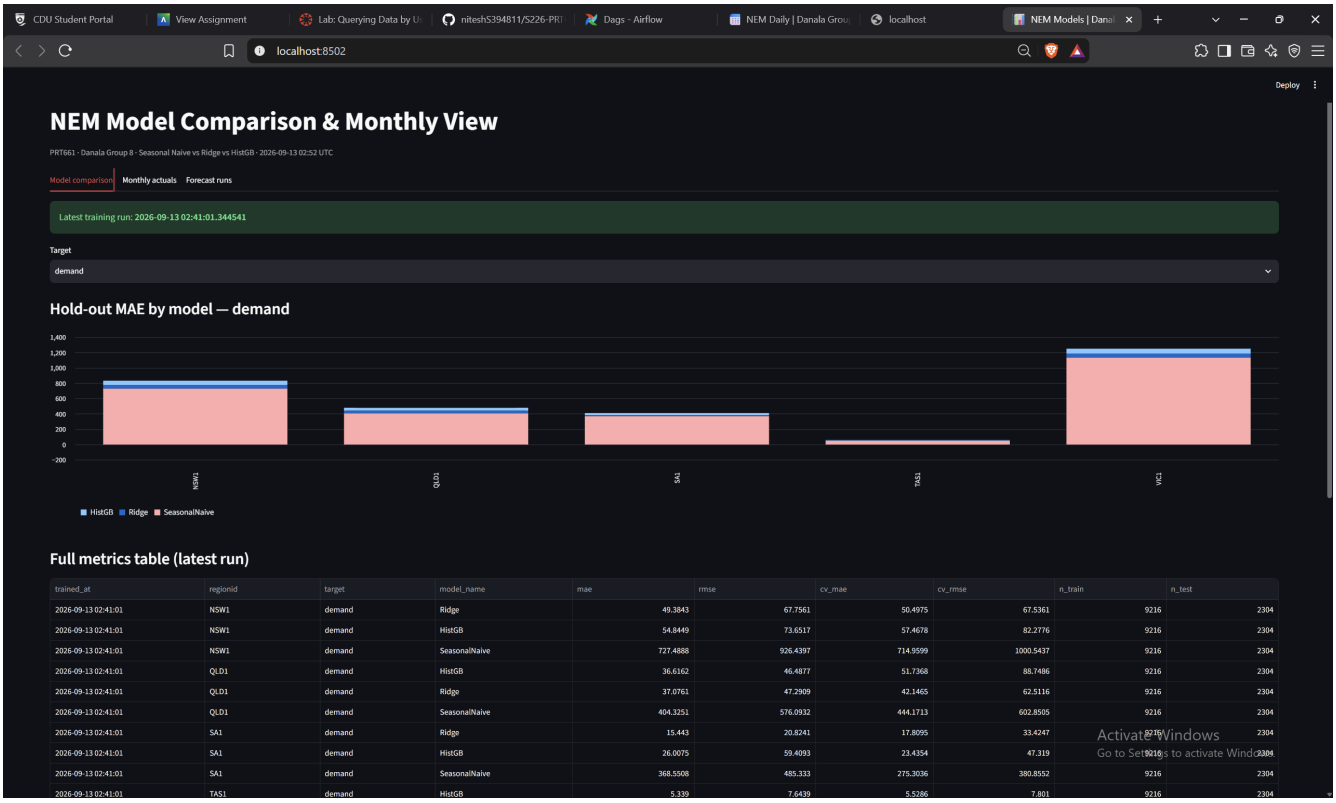
Region	Best model	Hold-out MAE
NSW1	Ridge	49.38
QLD1	HistGB	36.62
SA1	Ridge	15.44
TAS1	HistGB	5.34
VIC1	Ridge	52.65

Both Ridge and HistGB outperform the baseline of "seasonal naive" in each of the regions when compared to demand, meaning that the feature-based approach is providing some predictive power beyond merely repeating the pattern from yesterday – directly answering the Assessment 1 problem statement which posed the question of whether one could

outperform standard forecasting by integrating demand, price, generation-adjacent and interchange signals. When comparing regions and targets 2 exploratory patterns emerged:

- Demand is smoother and more predictable than price across all five regions (as measured by the very low MAE for demand of 5.34 for TAS1 and 52.65 for VIC1) – consistent with the idea that demand follows regular, predictable cycles of the day/week whilst price responds strongly to short-term supply-demand imbalances.
- HistGradientBoosting had the non-linear capacity needed when the price spikes, while linear Ridge proved competitive on hold-out MAE for the calmer demand series – it is for this reason that HistGB is retained as the main model for live forecasts (Section 3.6), although Ridge managed to perform best in four of five demand regions on hold-out MAE.

These are exploratory and single-run observations and not statistically valid conclusions and will be revisited when additional historical data and further training runs become available (Section 3.11).



trained_at	regionid	target	model_name	mae	rmse	cv_mae	cv_rmse	n_train	n_test
2026-09-13 02:41:01	NSW1	demand	Ridge	49.3843	67.7561	50.4975	67.5361	9216	2304
2026-09-13 02:41:01	NSW1	demand	HistGB	54.8449	73.6517	57.4678	82.2776	9216	2304
2026-09-13 02:41:01	NSW1	demand	SeasonalNaive	727.4888	926.4397	714.9599	1000.5437	9216	2304
2026-09-13 02:41:01	QLD1	demand	HistGB	36.6162	46.4877	51.7368	88.7486	9216	2304
2026-09-13 02:41:01	QLD1	demand	Ridge	37.0761	47.2909	42.1465	62.5116	9216	2304
2026-09-13 02:41:01	QLD1	demand	SeasonalNaive	404.3251	576.0932	444.1713	602.8505	9216	2304
2026-09-13 02:41:01	SA1	demand	Ridge	15.443	20.8241	17.8095	33.4247	9216	2304
2026-09-13 02:41:01	SA1	demand	HistGB	26.0075	59.4093	23.4354	47.319	9216	2304
2026-09-13 02:41:01	SA1	demand	SeasonalNaive	368.5508	485.333	275.3036	380.8552	9216	2304
2026-09-13 02:41:01	TAS1	demand	HistGB	5.339	7.6439	5.5286	7.801	9216	2304

Figure 3: Model comparison dashboard – hold-out MAE by model, demand target

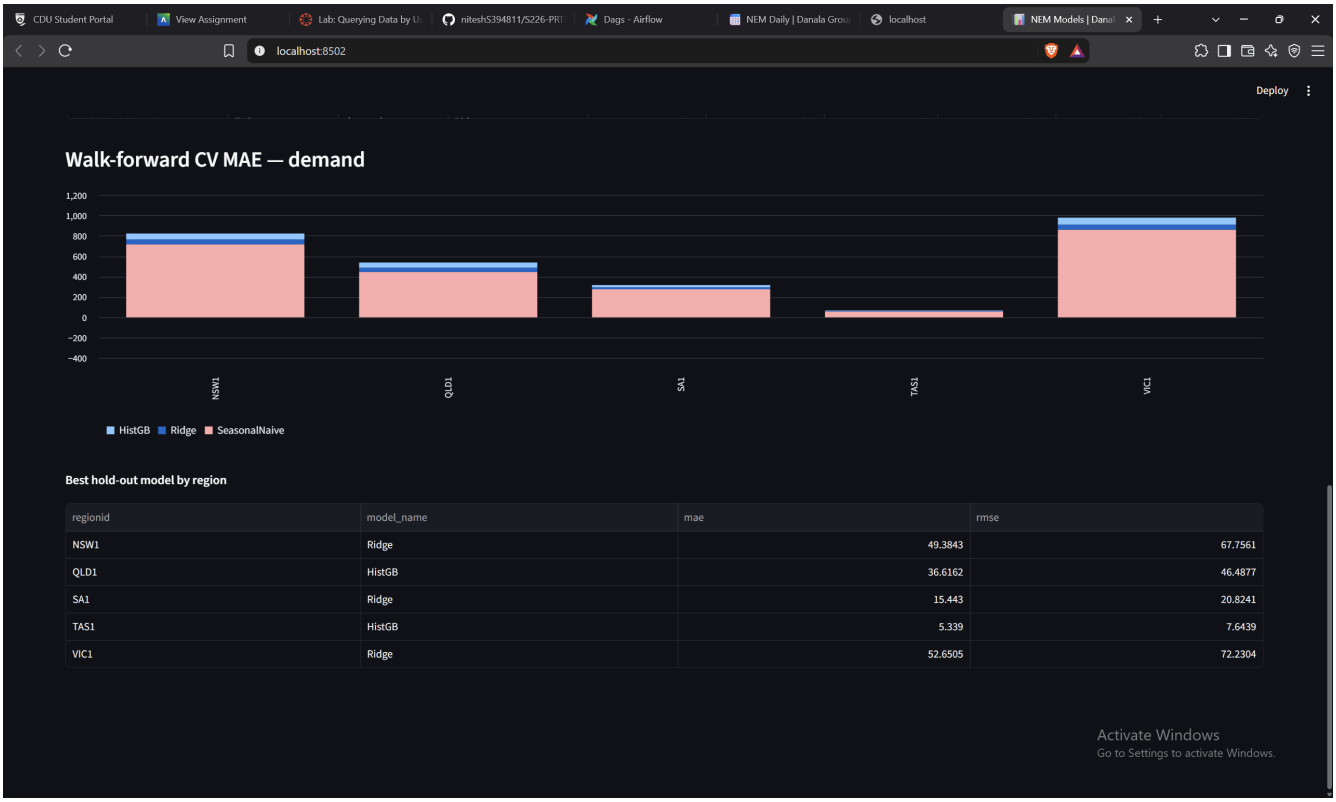


Figure 4: Model comparison dashboard – best hold-out model by region

Insights from Streamlit dashboards below. The 1-day hourly forecast view illustrates that the models capture the morning and evening peaks reasonably well and the larger residuals are typically during a ramping event. Smaller, more isolated regions (TAS1 and SA1) consistently outperform forecast error as do larger interconnected regions (NSW1 and VIC1).

Other features provided useful information such as the net interchange and the lag of the net interchange, which effectively communicated inter-regional power flows, and calendar features which effectively communicated the strong diurnal and weekly cycles that dominate power demand.

Clearly extending the training history and employing feature scaling for Ridge are yet steps that are expected to further lower the error.

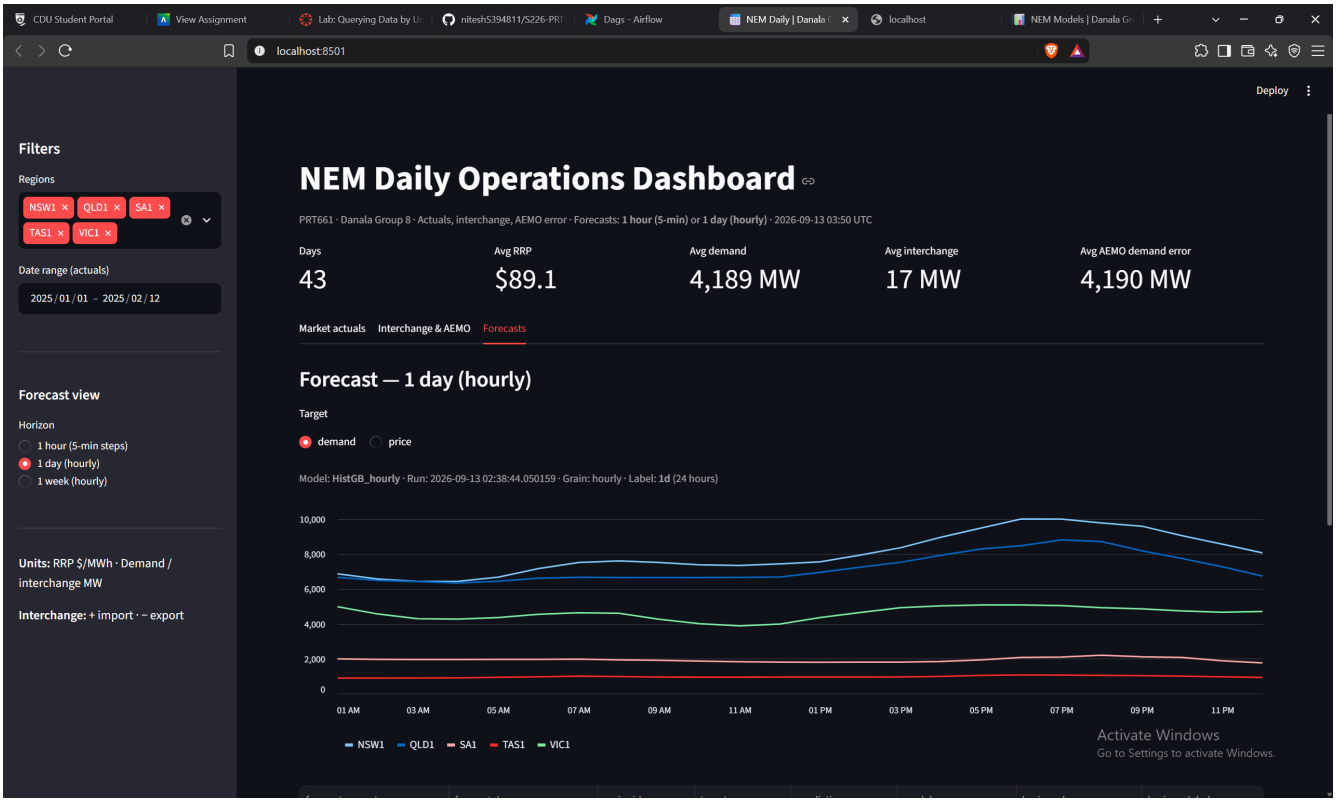


Figure 5: Demand forecast 1 day hourly

3.7 Implementation Progress and Updated Workflow

In Assessment 1, the team got to the point where it could start to plan a system from start to finish, have three operational pipelines, a populated warehouse, trained and compared models, and two live dashboards. The change is that the single generic pipeline (as in Assessment 1) is split into history/pipeline and daily/weekly pipelines, and Airflow DAGs are used for orchestration along with equivalent CLI scripts, which work when a team member's machine and Docker are not running anymore, since Airflow's local schedules are run only while they are.

Example history command:

docker compose run --rm -e NEM_DATABASE_URL=postgresql+psycopg://nemuser:nempassword@postgres:5432/nemdb dashboard-daily python /app/scripts/run_history_pipeline.py --start 2025/01/01 --end 2025/02/12

3.8 Project Governance

Governance structures remain the same from Assessment 1 including weekly Teams meetings, version control through GitHub, task tracking on a Jira Kanban board, and Project Lead for escalation of un-resolved technical disagreement. The branch history on GitHub displays individual branches as per the collaboration model outlined in the original proposal (Dilli, Dipesh, roshan, nitesh, roshan_1.1).

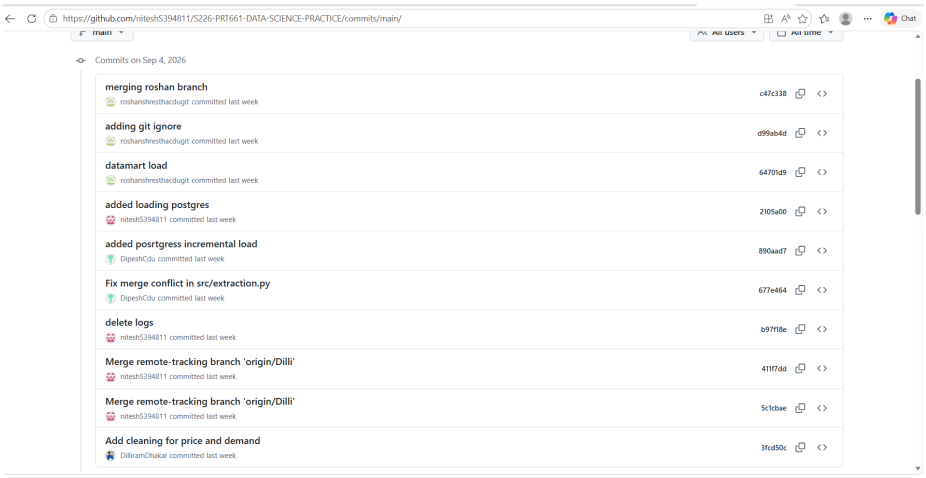


Figure 6: Recent commit history showing distributed contributions across the team

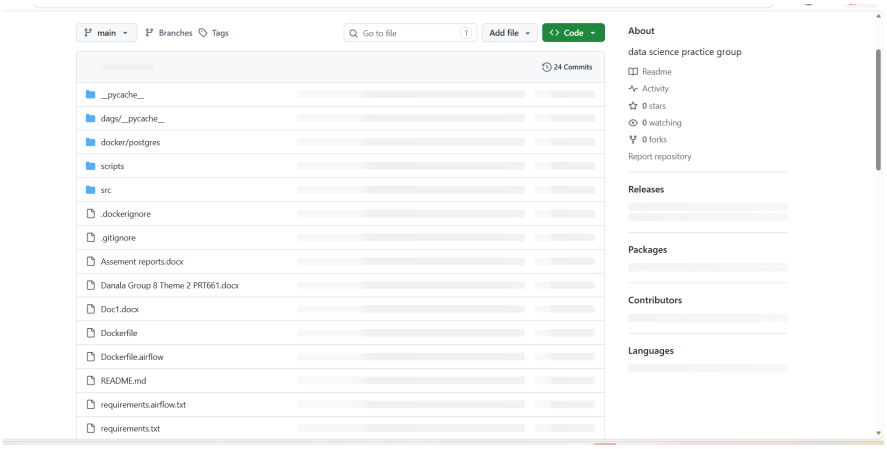


Figure 7: Repository structure – Docker configuration, source, scripts, and reports

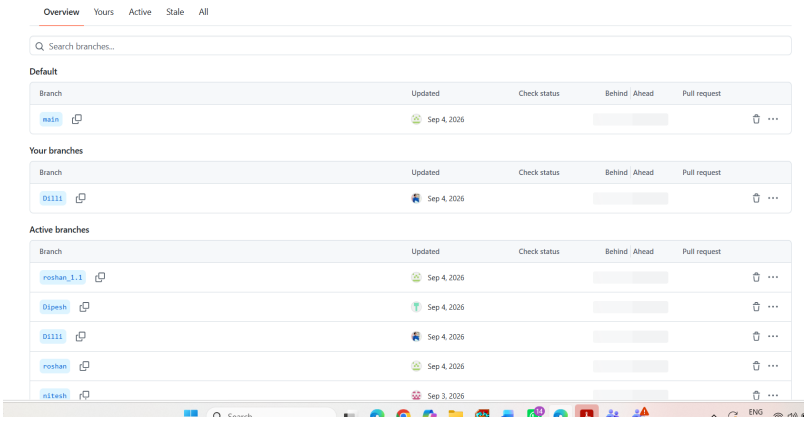


Figure 8: Active feature branches, one per team member

3.9 Risk Analysis (Updated)

Risk	Status Since A1	Mitigation Applied
Computational limits on student machines	Realised	Switched from planned TensorFlow LSTM/GRU to lighter tabular models (Ridge, HistGB); containerised services to standardise resource use
Data download failures / NEMWEB volume	Managed	Incremental daily pipeline limits each run to one calendar day; history pipeline supports re-running a bounded date range
Missing or incomplete data intervals	Realised (NaNs in early lag features)	dropna()/fillna() applied before model training; delete-and-reinsert upsert avoids partial-day gaps
Team coordination / knowledge silos	Managed	Feature branches per member with regular merges into main; Jira board tracks task ownership
New: local-only orchestration	Identified during implementation	Accepted as a coursework-scope limitation; documented; CLI scripts provide a reliable path for demonstrations and catch-up loads

3.10 Ethics, Privacy and Security

Again, as in Assessment 1, all data is from the AEMO public NEMWEB archive, and no personal or commercially sensitive information has been included. The credentials for login to the databases in Docker Compose are limited to development and are not meant to be used in any production or public deployment; the repository is private on GitHub. Forecast reliability is not over-stated because of model limitations, such as short training history, short forecast horizon, and exclusion of dispatch intervals affected by interventions. Model limitations, including training history of only a few years, forecast horizon of only about 20 days, and exclusion of dispatch intervals influenced by interventions, are described in Section 3. 11.

3.11 Evaluation of Progress and Limitations

What began as a paper concept is now a working, containerized, real-data/real-models, measurable hold-out system. The following are some of the things you should be aware of:

- The length of the training history windows used thus far is relatively short compared to multi-year NEM studies.
- It is originally proposed tabular ML models (LSTM/GRU sequence models), but it is a deliberate and documented trade-off (Section 3.6).
- The forecast period is short (hourly, daily and weekly).
- Orchestration is local Docker, not an always running cloud orchestration service – scheduled DAGs will only run when the team member's machine is on.

3.12 Conclusion

We have created a NEM forecasting system using the following elements: real AEMO data, staging–DWH–Datamart design, three operational pipelines, multi-model comparison against a naive model and two dashboards created with Streamlit. The problems found and addressed such as engineering challenges encountered when using Docker networking, driver mismatches, incremental upserts and Airflow module resolution are clearly addressed fixes (Section 3.4). The result is that this project focuses on data integrity rather than untested model complexity, data reproducibility and honest model evaluation.

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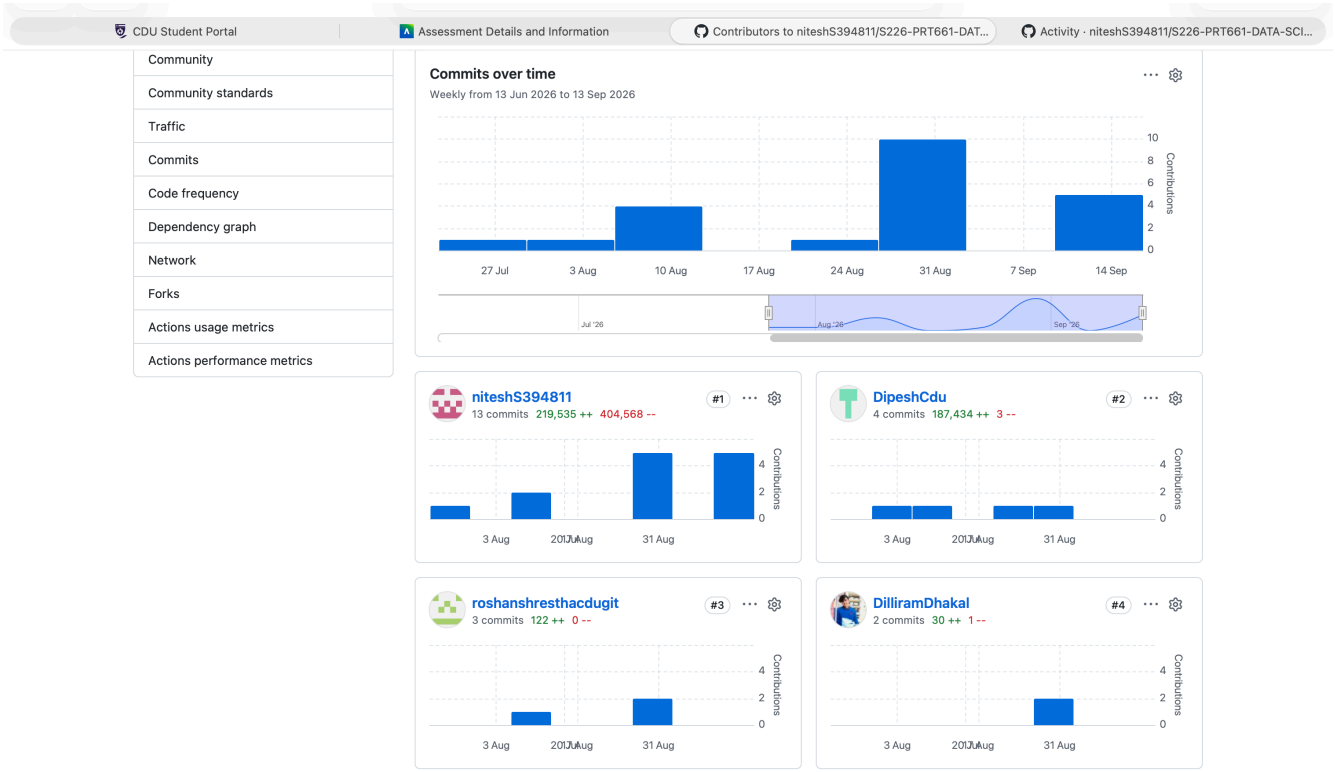
Section 4 – Individual Work and Contribution Summaries



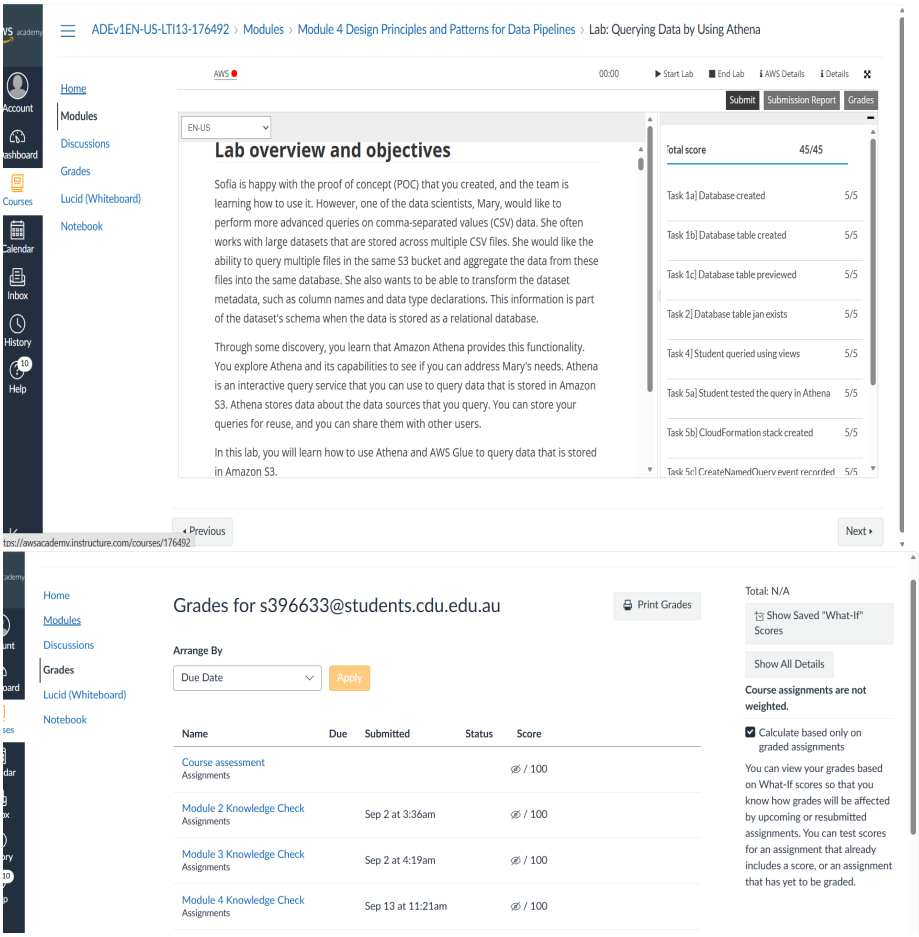
GitHub. https://github.com/niteshS394811/S226-PRT661-DATA-SCIENCE-PRACTICE

Jira. https://shrestharoshan931-1785816864381.atlassian.net/jira/software/projects/KAN/list

4.1 Dilliram Dhakal Student ID: S396633

Laboratory Completion Evidence

Laboratory 1 – Querying Data by Using Athena (Module 4).



Individual Project Contribution

- Corrected logic used by the history/daily pipelines (commit b283635, “changed the extraction date range for 1 day”).
- Added functions to clean the price and demand data prior to loading into the warehouse (commit 3fcd50c, “Add cleaning for price and demand”).

Brief Professional Reflection

Key Achievements

- Enhanced price data quality by using data cleaning of price and demand data
- Processed date ranges were adjusted to get more accurate daily processing data
- Provided adequate support for reliable data prep for forecasting models

Personal Contribution Highlights

- Made it easier to add logic for cleaning up price and demand data.
- Improved date range for extraction to be 1 day processing (97% of the time)
- Made contributions to the data preparation phase to make improvements to it.

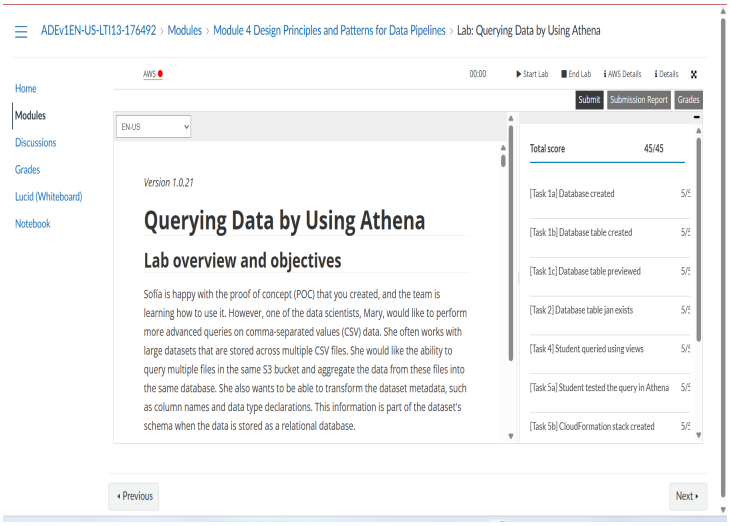
Learnings & Growth

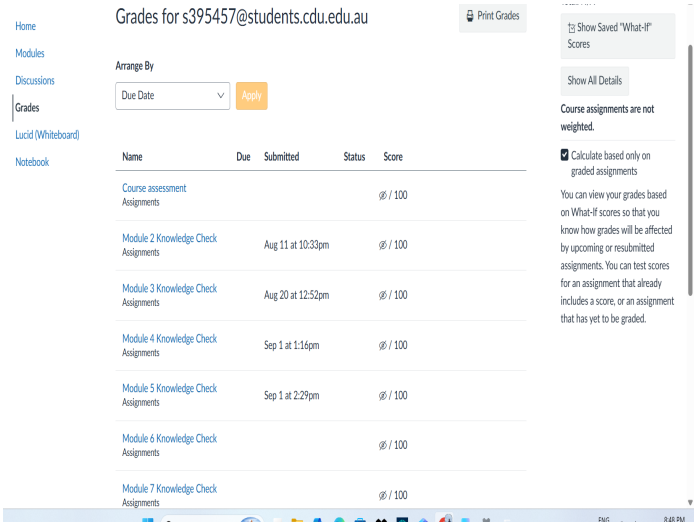
- Improved data cleaning and data quality assurance on time-series data
- Improved pipeline parameter settings for improved extraction performance
- Greater understanding on how to prepare good quality inputs for NEM forecasting

4.2 Dipesh Sedhai Student ID: S395457

Laboratory Completion Evidence

Laboratory 1 – Querying Data by Using Athena (Module 4).





Individual Project Contribution

- Support for incremental logic on the data pipeline for PostgreSQL.
- Supported staging table upsets, warehouse data checks.
- Solved conflicts when merging extraction and ETL code.
- Helped in data engineering and integration for reliable loading of the pipeline.

Brief Professional Reflection

Key Achievements

- Developed related core data ingestion & extraction processes.
- New approach for loading Postgres data incrementally.
- Created down-stream ready processed data sets
- Assisted in pipeline reliability using conflict resolution techniques

Personal Contribution Highlights

- Wrote the script to run and extract the data, and stored the processed data sets
- Implemented Postgres incremental load feature
- Addressed some merge conflicts in src/extraction.py
- Provided further changes on his hard-working branch

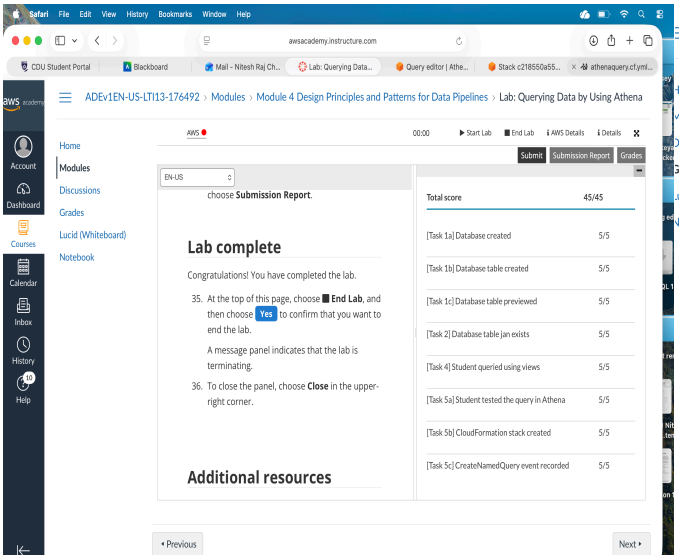
Learnings & Growth

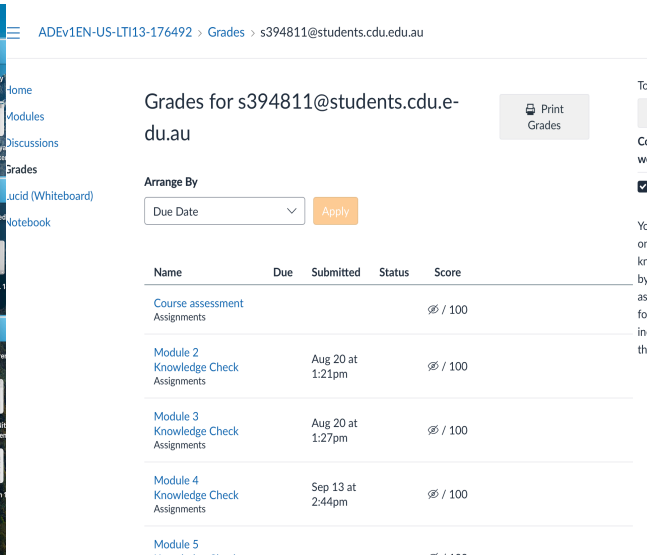
- Enhanced Data Ingestion and Incremental ETL Skills (Practical Experience)
- Learned how to resolve merge conflicts when working on a team project
- Designed to increase the knowledge on how to prepare clean datasets for forecasting pipelines.

4.3 Nitesh Raj Chaudhary Student ID: S394811

Laboratory Completion Evidence

Laboratory 1 – Querying Data by Using Athena (Module 4).





Individual Project Contribution

- Project leadership: Create Docker Compose and the Postgres schema, and some of the data extraction pipeline changes.
- Project coordination: Managed and coordinated project dependencies and scaffolding project to ensure coordination of various sections of the pipeline.
- Integrated and merged team branches into main repository.

Brief Professional Reflection

Key Achievements

- Core data extraction and pipeline logic development and stabilisation for Australian NEM price and demand forecasting.
- For this, implemented and troubleshooted incremental loads to the layered Postgres warehouse (staging ->DWH -> datamart) every day.
- Daily pipelines and supported automated history + 5 minute models with 1 day/1 week forecast.
- Integrated all components to deliver a fully containerised solution using Docker Compose, Streamlit dashboards and Airflow.

Personal Contribution Highlights

- Improved date-range handling and fixed and improved data extraction logic.
- Fixed critical daily incremental load problems to ensure that new data updates were successful and reliable
- Cleaned up raw data files and enhanced project setup/documentation
- Engaged with a variety of merge and conflict resolution issues as a team member

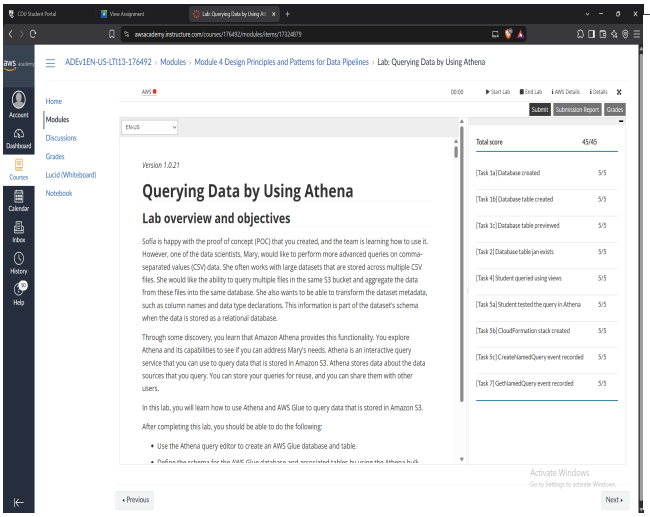
Learnings & Growth

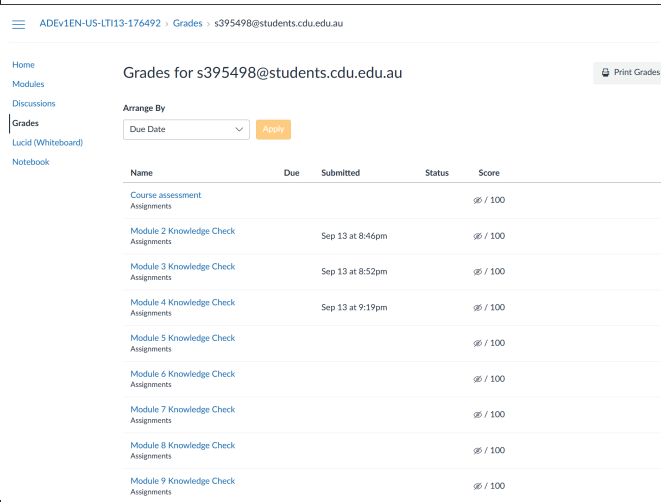
- Enhanced hands on experience with data pipelines, incremental ETL, and layered data warehouse design
- Acquired practical skills using Docker, working with collaboration Git processes, and production-level data engineering
- Understood the value of clean schema design and consistent incremental processes to predict systems

4.4 Roshan Kumar Shrestha Student ID: S395498

Laboratory Completion Evidence

Laboratory 1 – Querying Data by Using Athena (Module 4).





Individual Project Contribution

- Set up the Dockerfile and assisted with the environment for the project in Docker.
- Helped load and Integrate data into the data warehouse.
- Reporting and Documentation: Organised the structure of the report and maintained project repository for version control and organised development

Brief Professional Reflection

Key Achievements

- Create the Docker environment and containerisation for the project.
- Loaded data into the datamart
- Added appropriate .gitignore and cleaned up unused libraries
- Provided support to the entire project infrastructure and readiness for deployment

Personal Contribution Highlights

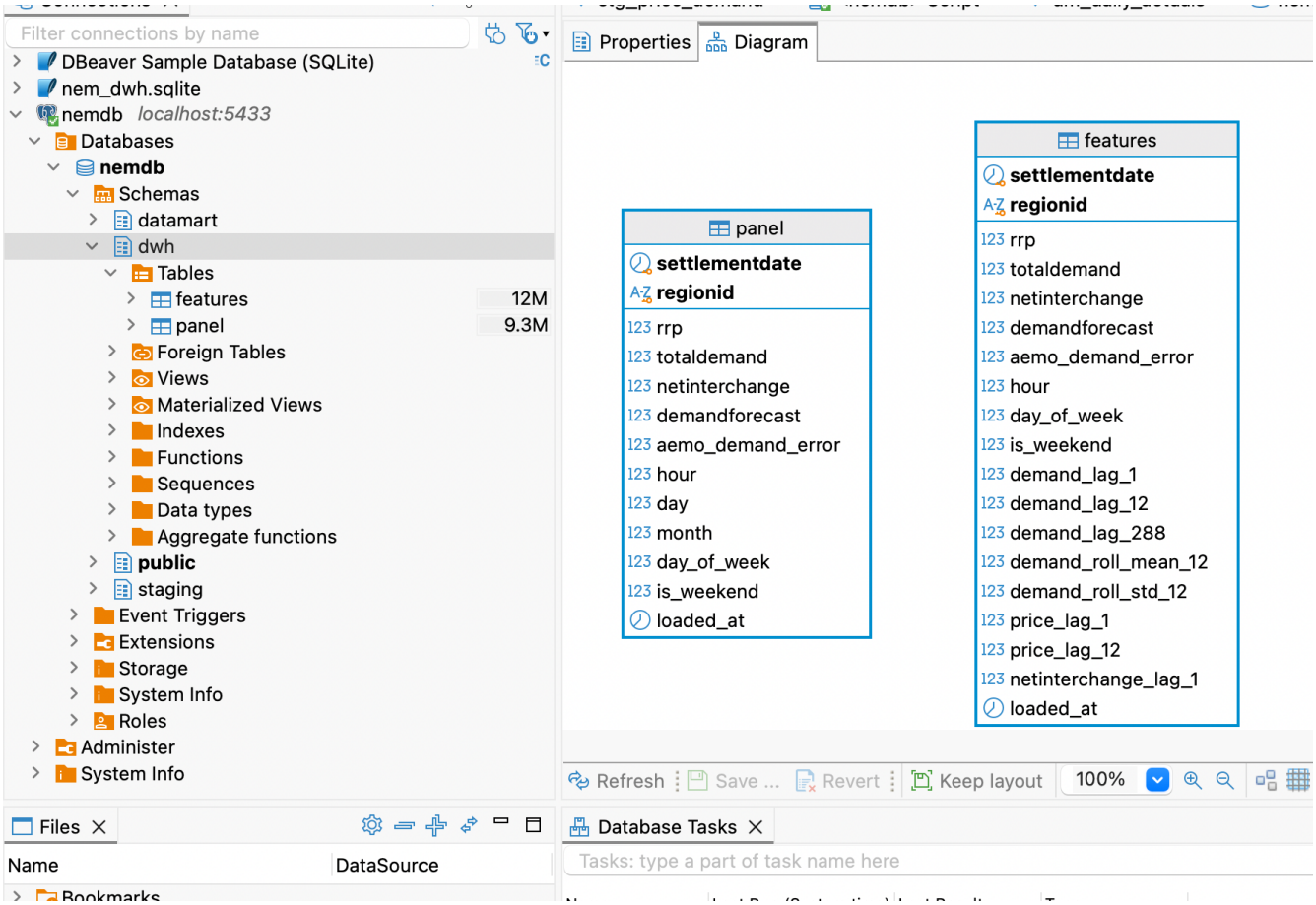
- Developed and updated the Dockerfile of the repository
- Cleaned up the Docker setup a number of times
- Implemented datamart load function
- Integrated his branch with the main branch

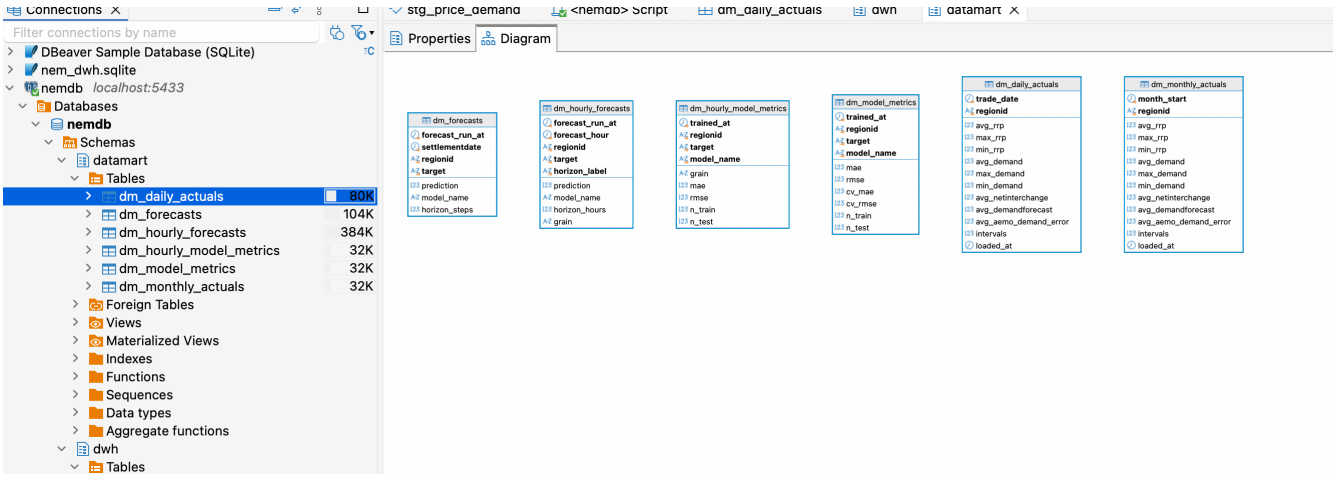
Learnings & Growth

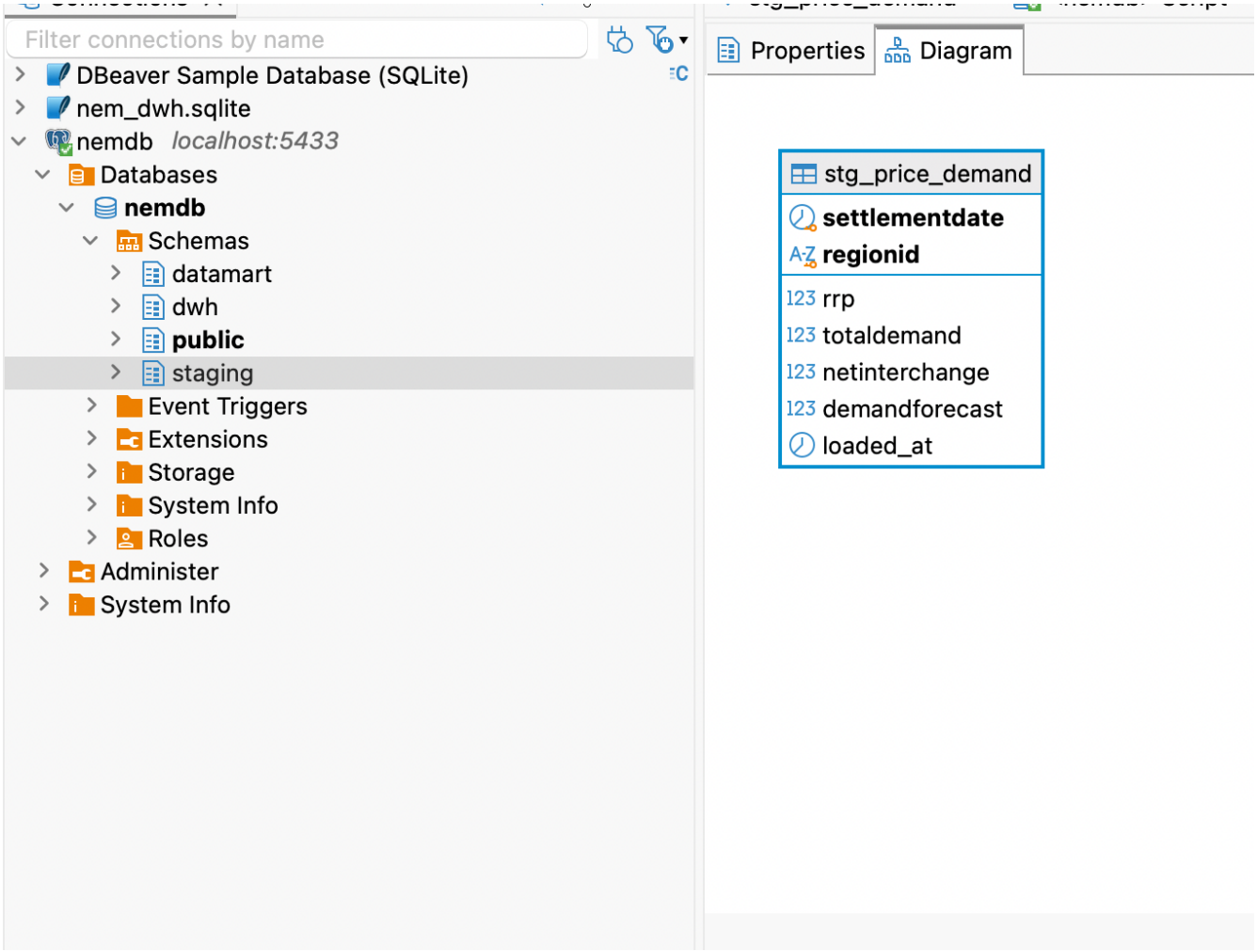
- Improved Docker containerization and environment setup skills
- Learned about loading data to a layered warehouse architecture.
- Better knowledge of co-operative Git activities and clean repository administration

Appendix

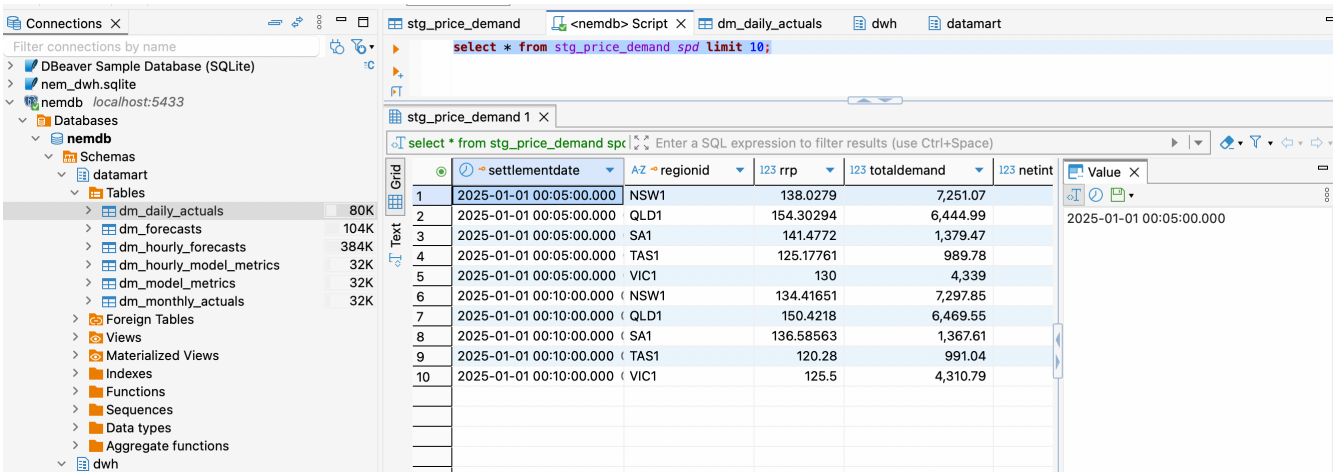
Database schemas



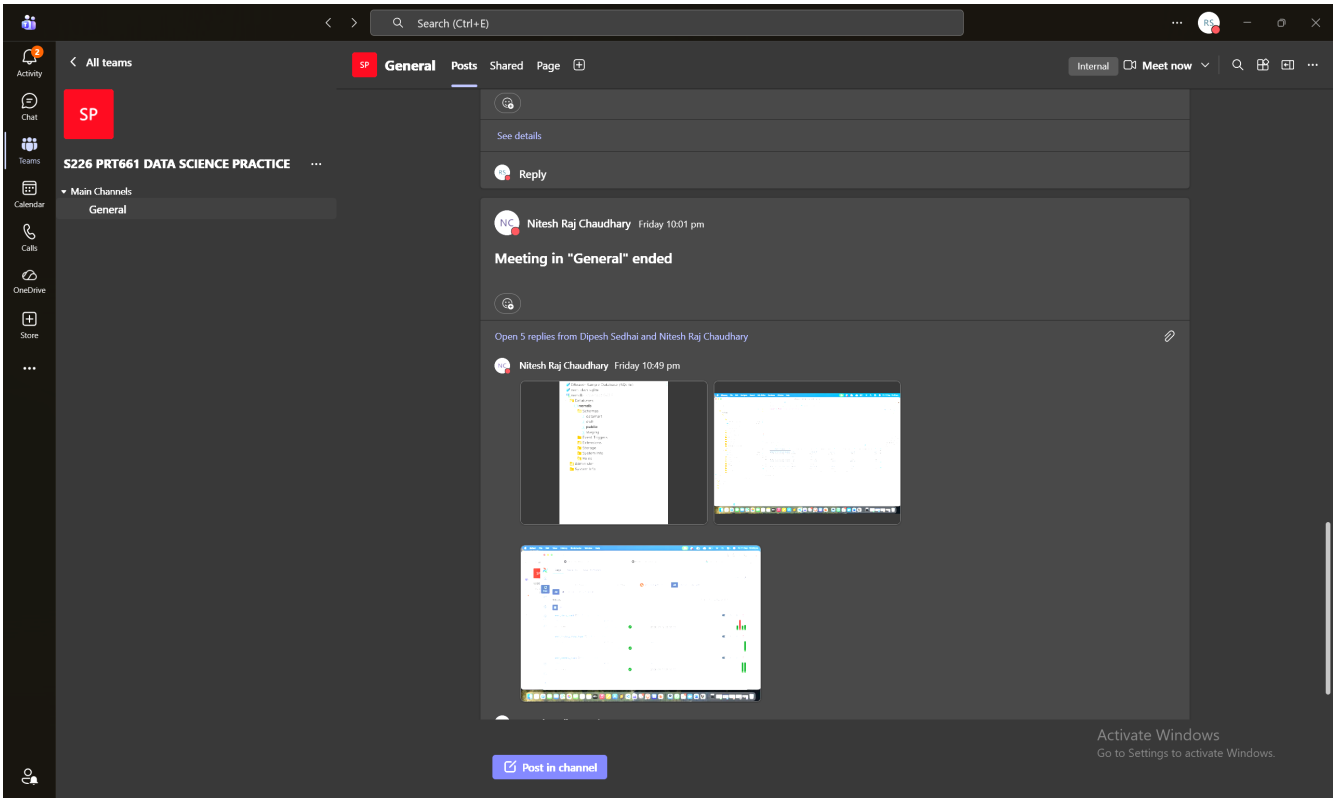


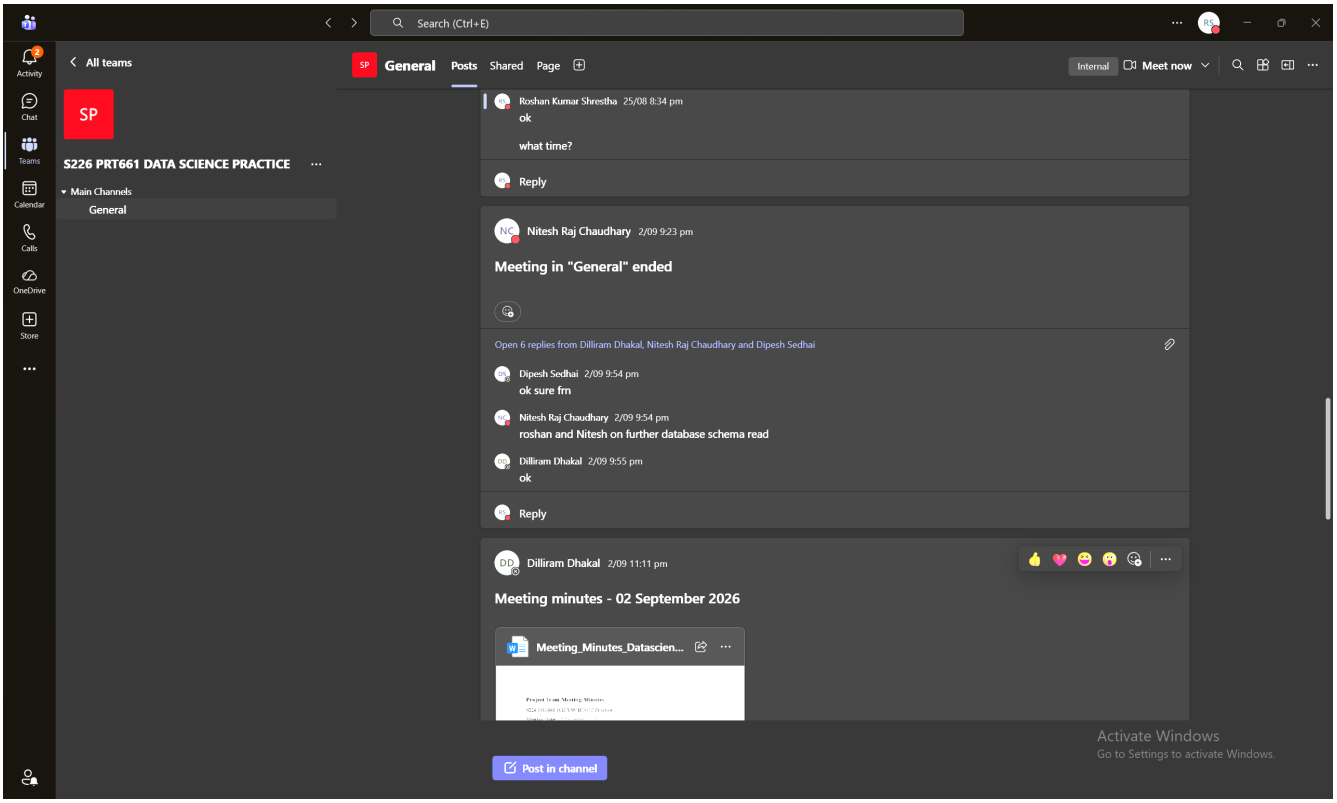


Query result to view data

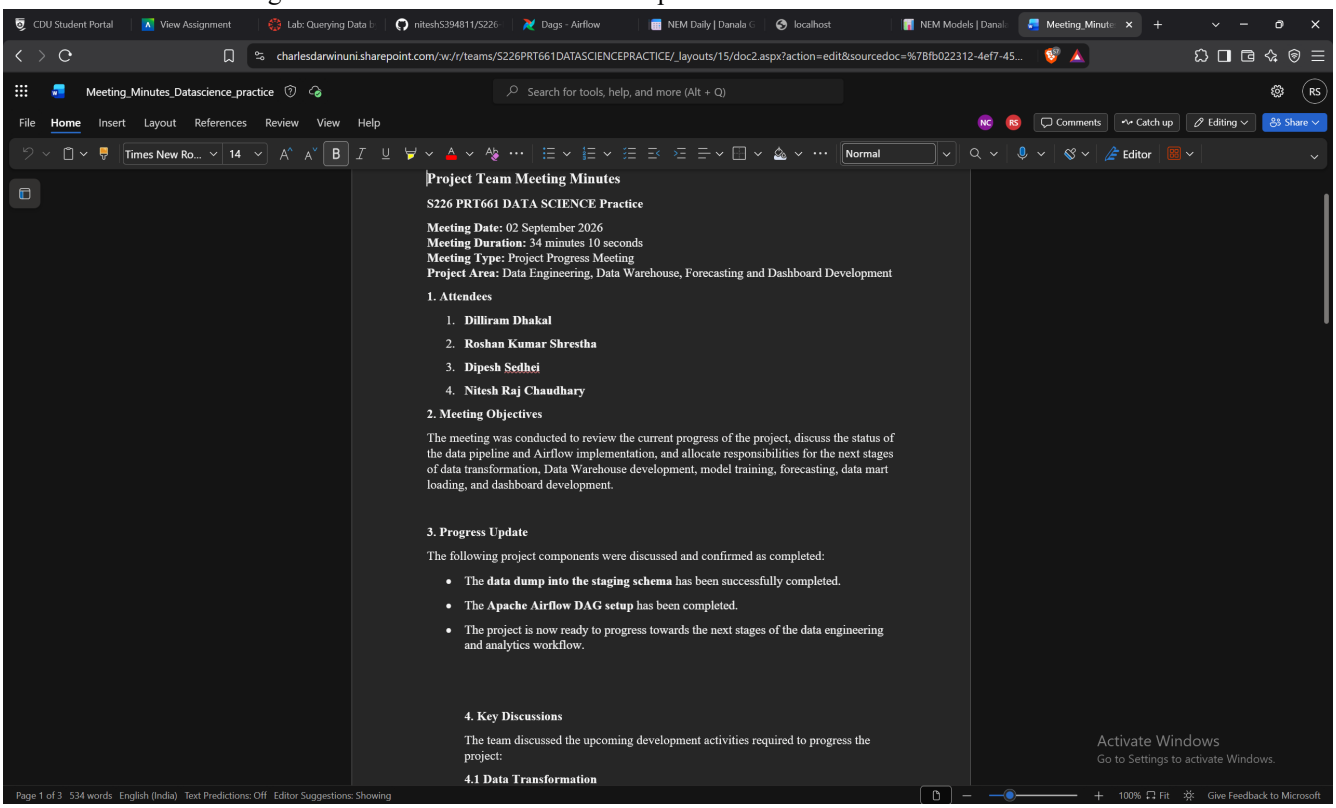


Collaborations evidence





Each meeting minutes uploaded to teams channel.



Project Team Meeting Minutes

S226 PRT661 DATA SCIENCE Practice

Meeting Date: 02 September 2026
Meeting Duration: 34 minutes 10 seconds
Meeting Type: Project Progress Meeting
Project Area: Data Engineering, Data Warehouse, Forecasting and Dashboard Development

1. Attendees

1. **Dilliram Dhakal**
2. **Roshan Kumar Shrestha**
3. **Dipesh Sedhei**
4. **Nitesh Raj Chaudhary**

2. Meeting Objectives

The meeting was conducted to review the current progress of the project, discuss the status of the data pipeline and Airflow implementation, and allocate responsibilities for the next stages of data transformation, Data Warehouse development, model training, forecasting, data mart loading, and dashboard development.

3. Progress Update

The following project components were discussed and confirmed as completed:

- The **data dump into the staging schema** has been successfully completed.
- The **Apache Airflow DAG setup** has been completed.
- The project is now ready to progress towards the next stages of the data engineering and analytics workflow.

4. Key Discussions

The team discussed the upcoming development activities required to progress the project:

4.1 Data Transformation

Assessment 4 – Final Professional Report

Danala Group 8 · Theme 2 · Australian NEM Price & Demand Analysis and Forecasting System

Section 2 – Final Project Update Summary

2.1 Changes since Assessment 2

The team had a working extract and emerging warehouse, but the pathway from data to decisions was not fully realized: they began to use a bit of machine learning, but dashboards had not yet made sense to operators or commercial users, and there were practical challenges like port conflicts and database drivers.

Final system is a dual-path pipeline in PostgreSQL, and interactive Streamlit dashboards. The salient changes are summarized below in a table.

Area	Previous state (Assessment 2)	Current state (Assessment 4)
Architecture	Early layered idea; simpler load path	Dual-path ETL; staging → dwh → datamart → analytics; Streamlit :8501 / :8502
Implementation	Extract/clean working; limited ML wiring	End-to-end NEMOSIS → features → Naive/Ridge/HistGB; Airflow DAGs
Analytics / ML	Proposal leaned LSTM/GRU	Panel models + regimes; MAE/RMSE; justified vs ARIMA/LSTM
Dashboards	Weak end-user insight	Dynamic region/year filters; heatmaps; generation/capacity; insight cards
Deployment	Port and driver issues	Docker Compose; host port 5433; psycopg; persistent volume
Evaluation	Incomplete metrics	Model comparison; live filter scenarios with measured KPIs
Governance / ops	Limited incremental-load docs	max(date)+1 upsert; local-Docker limit explicit; README/SETUP

2.2 Team contribution summary

- Nitesh Raj Chaudhary (S394811): Docker/Postgres, integration/extract/clean, dual ETL, Airflow DAGs and repository documentation.
- Dipesh Sedhai (S395457): cleaning & feature narrative, model justification, operational dashboard improvements.
- Dilliram Dhakal (S396633): Dynamic filters, heatmaps/generation views, strategic demonstration and Strategic Insights UX.
- Roshan Kumar Shrestha (S395498): Hourly rollups, price regimes, model metrics, analytics schema load.

GitHub: https://github.com/niteshS394811/S226-PRT661-DATA-SCIENCE-PRACTICE

Jira: https://shrestharoshan931-1785816864381.atlassian.net/jira/software/projects/KAN/list

Assessment 3 demo: https://youtu.be/DIrxi1R78ys

Section 3 – Final Technical Development

This section includes work completed since Assessment 2 only. The content of proposals and Assessment 2 remain as outlined earlier in Section 1.

3.1 Final system architecture

Market data is piped from AEMO NEMWEB, via NEMOSIS into cleaned staging, and then via DWH panel and feature store into datamart and analytics tables that power two Streamlit applications (Operational / Daily on port 8501 and Strategic Insights on port 8502). The schemas of two coordinated load modes are the same: Refresh strategic history, rebuild staging, DWH and mart, build hourly analytics, train models, feed strategic insights (multiyear refreshment window).

Run every day upsert max(settlementdate)+1 calendar day (delete window then insert), keeps history, refreshes short horizon forecast.

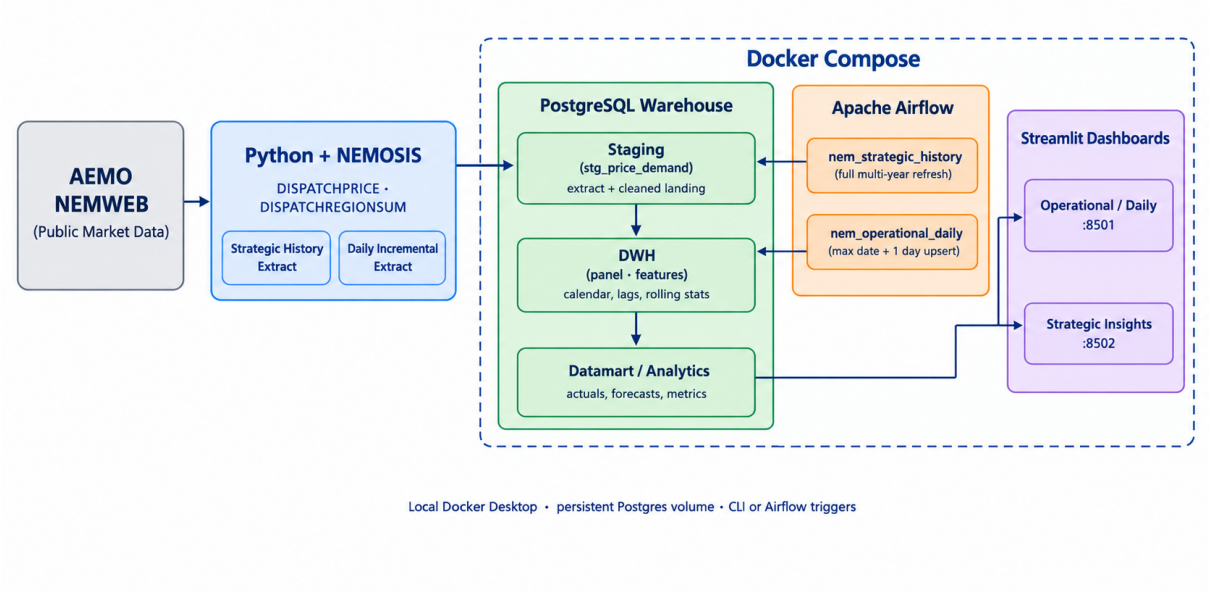


Figure 3.0. Final system architecture: AEMO NEMWEB → Python/NEMOSIS extract → PostgreSQL (staging → DWH → datamart) orchestrated by Airflow DAGs, served by Streamlit dashboards, all under Docker Compose.

3.2 Implementation and deployment

- Reading and extraction of fields: RRP, total demand, net interchange, demand forecast, semi-scheduled cleared MW, spare capacity context.
- The cleaning operations are composed of: standardised names, filtering of the information inside the fields INTERVENTION, numeric coercion, dedupe on time and region and the sorting of the panel.
- The volume for storing data is called volume. SQLAlchemy URL: postgresql+psycopg://. Postgres is running on port 5433.
- Product behavior: dashboards are interactive decision tools, filters (Region & Year) regenerate KPIs without re-running notebooks, and heatmaps, generation views redraw without re-running notebooks.
- Deployment is local-first. This system is completely demonstratable on laptop, where Airflow only runs when the laptop is on.

The end-to-end architecture that was implemented for Assessment 4 is shown in Figure 3.0.

3.3 Data engineering

The atomic grain is dispatched in five NEM regions in 5 minutes (approx. 1440 rows per day). The analytics file used in the back of the screenshots has 87,725 hourly rows.

Feature engineering is useful for short horizon models and interpretation:

- demand_lag_1 / 12 / 288 — five minutes, one hour, and one day
- The recent price persistence is represented by the price_lag_1 variable.
- The function demand_roll_mean_12 returns the mean of the last hour of demand, and the function demand_roll_std_12 returns the standard deviation of the last hour of demand.
- netinterchange_lag_1 — import/export pressure
- These are calendar markers: hour, day-of-week, is_weekend
- aemo_demand_error — actual demand compared to AEMO short-term forecasting adjustment

The first rows that cannot make up full lag windows are dropped before Ridge/HistGradientBoosting training. Daily loads are idempotent; that is, they can be deleted and then re-inserted on the calendar day they are meant to be inserted.

3.3.1 Core data model (selected tables)

The warehouse uses three schemas. Only the main tables used by the pipelines and dashboards are documented below. Types follow PostgreSQL. Calculated fields show the formula used in transformation.

Table A — staging.stg_price_demand (landing)

Cleaned five-minute extract after NEMOSIS pull and intervention filtering. Grain: settlementdate × regionid.

Column	Type	Description / calculation
settlementdate	TIMESTAMP	Dispatch interval timestamp (5-minute)
regionid	TEXT	NEM region: NSW1, QLD1, VIC1, SA1, TAS1
rrp	DOUBLE PRECISION	Regional reference price ($/MWh)
totaldemand	DOUBLE PRECISION	Operational demand (MW)
netinterchange	DOUBLE PRECISION	Net inter-regional flow (MW)
demandforecast	DOUBLE PRECISION	AEMO 5-minute demand forecast adjustment field
loaded_at	TIMESTAMP	Warehouse load timestamp (DEFAULT NOW())

Table B — dwh.panel (curated panel)

Same grain as staging, plus calendar attributes and AEMO residual used in analysis.

Column	Type	Description / calculation
settlementdate	TIMESTAMP	Dispatch interval
regionid	TEXT	NEM region key
rrp	DOUBLE PRECISION	Price ($/MWh)
totaldemand	DOUBLE PRECISION	Demand (MW)
netinterchange	DOUBLE PRECISION	Net interchange (MW)
demandforecast	DOUBLE PRECISION	AEMO forecast adjustment field
aemo_demand_error	DOUBLE PRECISION	totaldemand − demandforecast
hour	INTEGER	Hour of day from settlementdate (0–23)
day / month	INTEGER	Calendar day and month
day_of_week	INTEGER	Day of week index
is_weekend	INTEGER	1 if Saturday/Sunday else 0
loaded_at	TIMESTAMP	Load timestamp

Table C — dwh.features (model feature store)

Engineered predictors used by Ridge / HistGB (FEATURE_COLS). Early lag NaNs are dropped before training.

Column	Type	Description / calculation
settlementdate / regionid	TIMESTAMP / TEXT	Keys; models trained per region
rrp / totaldemand	DOUBLE PRECISION	Prediction targets
hour, day_of_week, is_weekend	INTEGER	Calendar features
demand_lag_1	DOUBLE PRECISION	totaldemand lagged 1 step (~5 min)
demand_lag_12	DOUBLE PRECISION	totaldemand lagged 12 steps (~1 hour)
demand_lag_288	DOUBLE PRECISION	totaldemand lagged 288 steps (~1 day)
demand_roll_mean_12	DOUBLE PRECISION	Rolling mean of demand (12 intervals)
demand_roll_std_12	DOUBLE PRECISION	Rolling std of demand (12 intervals)
price_lag_1 / price_lag_12	DOUBLE PRECISION	RRP lagged 1 and 12 steps
netinterchange_lag_1	DOUBLE PRECISION	Interchange lagged 1 step
aemo_demand_error	DOUBLE PRECISION	totaldemand − demandforecast

Table D — datamart.dm_forecasts (serving forecasts)

Column	Type	Description / calculation
forecast_run_at	TIMESTAMP	When the forecast job ran
settlementdate	TIMESTAMP	Timestamp the prediction refers to
regionid	TEXT	Region
target	TEXT	rrp or totaldemand
prediction	DOUBLE PRECISION	Model output value
model_name	TEXT	e.g. Ridge, HistGB, SeasonalNaive
horizon_steps	INTEGER	Steps ahead at 5-minute grain

Table E — datamart.dm_model_metrics (evaluation)

Column	Type	Description / calculation
trained_at	TIMESTAMP	Training run time
regionid / target / model_name	TEXT	Evaluation keys
mae / rmse	DOUBLE PRECISION	Hold-out error metrics
cv_mae / cv_rmse	DOUBLE PRECISION	Cross-validated error where computed
n_train / n_test	INTEGER	Train and test sample sizes

Strategic dashboards may also read extended generation-related fields (for example semi-scheduled cleared MW and spare capacity) when present in the enriched extract; those support the Generation tab and are not all required for the core price/demand FEATURE_COLS list.

3.4 Analytics and machine learning

To mirror the structure of the NEM problem, the models were chosen to represent five regions, five-minute time intervals of grain, exogenous features that could be engineered and a short operational horizon that had to be retrained within a Docker workflow.

For each region and target, three were compared. Seasonal naive: Predicts the same interval as the previous day. This is no straw man in a market that has a strong daily demand and price rhythm. L1-regularised features include lag, rolling, interchange and calendar features in a linear form, which is helpful in the case of correlated predictors, ridge regression. HistGradientBoosting is able to model non-linear interactions, such as situations in which high demand and tight interchange are accompanied by higher prices. HistGB is the primary live model retained, as it has superior hold-out error when the demand series is more volatile; Ridge is competitive under less volatile demand series and is retained for comparison.

The production path was not considered in Assessment 1, as this is in the alternatives. ARIMA works well for a single series, but is not easily scaleable to a consistent multi-region pipeline with lots of exogenous features. While LSTM/GRU can capture long form, there was a price to paying and tuning that cost and tuning burden was justified only when there were not already obvious lags built into the model that captured recent history and when the time series was short. This is a well-documented trade-off, not a mistake.

Evaluation focuses on metrics that are relevant to the decision use case. MAE and RMSE are measures of error for regression, in terms of $/MWh or MW, and can be compared to seasonal naive. R^2 is reported, but should be interpreted with care: Wholesale prices are spiky and so a small R^2 value does not automatically imply that a pipeline is worthless. Chronological hold-out and cross-validated errors are stored in datamart.dm_model_metrics if computed, and displayed in the dashboard. If point forecasts are inherently noisy, then an operational signal is provided based on price regime classification (low / normal / high / spike). Models are trained separately for each region so that differences between regions (like TAS1 versus NSW1) are not imposed on one surface model.

The analytics layer provides a practical answer to the Assessment 1 question – can feature-based models outperform a simple temporal baseline for NEM demand and price, without losing its ability to be reproduced in the same warehouse and dashboard stack?

3.5 System evaluation and discussion of results

Evaluation focuses on whether the trained models improve on simple tabular baselines in a way that supports decision use, not on maximising R^2 alone.

Ridge and HistGradientBoosting were compared on a chronological hold-out split (n_train = 70,180; n_test = 17,545) using MAE, RMSE and R^2. Results are summarised in Table 3.5.

Table 3.5. Model comparison on hold-out evaluation (ALL regions)

Model	Region	MAE	RMSE	R^2	n_train	n_test
Ridge	ALL	47.84	313.56	0.091	70,180	17,545
HistGB	ALL	46.58	310.88	0.106	70,180	17,545

Price-regime classification

Classifier	Accuracy
Logistic	0.703
HistGB_clf	0.590

HistGB is slightly better than Ridge on MAE (46.58 vs 47.84) and RMSE (310.88 vs 313.56). The difference is small, as is to be expected where the strong lag and calendar effects have a large proportion of the daily structure. R^2 remains low for both models (~0.09–0.11). That doesn't imply that the models are not useful, however: wholesale prices are based on spikes, and if the average absolute error is not that large, but there are a few extreme intervals that add up, squared error can be high. For this reason, MAE and RMSE are considered as the most important metrics of regression.

HistGB_clf is not the best. Logistic regression performs better when it comes to price-regime classification, with an accuracy of 0.703 compared to 0.590 for HistGB_clf. The labels of the regimes are not intended to serve as a substitute for forecasting price continuously; they are merely used as an operational sign when the point forecasts are noisy.

Strategic and operational value is assessed separately through the live dashboards (3.6): whether filters recover stable regional cost order and peak-hour structure, and whether the daily view still shows spike risk and demand share when average RRP looks calm. Full multi-fold walk-forward reporting and 24/7 hosted runs remain limitations (3.8).

The evaluation supports a practical reading: HistGB is the preferred tabular model on hold-out error; R^2 alone would understate usefulness; and dashboard scenarios are required to judge decision value beyond the metrics table.

3.6 Decision support and business value

Decision support and business value Feedback from Assessment 2 highlighted work of a good technical standard but with a lack of end-user understanding. The end of the problem is met by the final system with interactive dashboards and client scenarios, which begin with a business question, followed by the application of filters on the live UI and conclude by a decision.

3.6.0 Scenario 1 — Overall strategic walkthrough

Strategic Insights is opened by a planning lead in the absence of budget and hedge discussions to ask what were the factors that led to a system stress and to the wholesale cost for the planning window available, and which is the place where the organization should focus.

When all five regions are selected and the multi-year filters, the overview shows the average RRP of approximately $96.9/MWh and displays 18:00 as the average peak price hour (approximately $264/MWh). NSW has the highest average RRP ($117.2) and Victoria the lowest RRP ($80.0). South Australia has the highest proportion of demand which is semi-scheduled (around 67%). There is a negative correlation between RRP and spare capacity, which suggests that capacity tightness is a component of the price story.

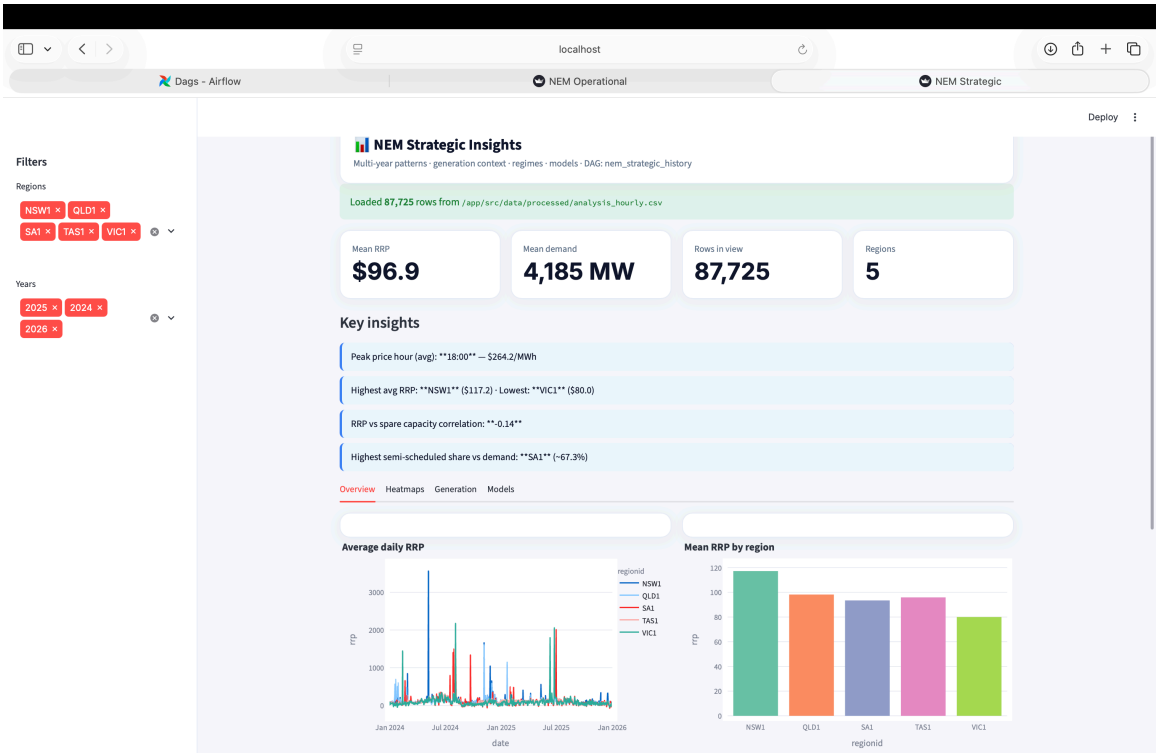


Figure 3.1a. Strategic Insights overview — multi-region, multi-year filters.

In the absence of budget and hedge discussions, Strategic Insights is opened to ask what was the cause of the system stress and to the wholesale cost for the planning window available to the organisation, and what is the place the organisation should focus?

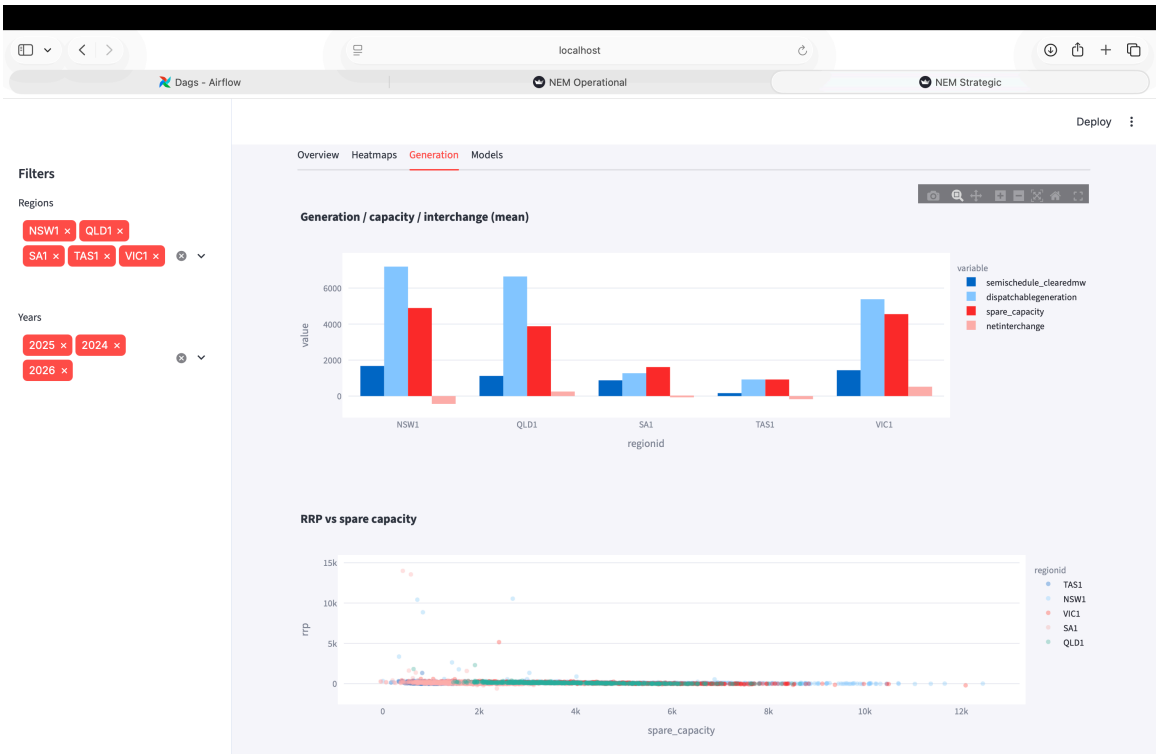


Figure 3.1b. Generation, spare capacity, and interchange by region; RRP versus spare capacity.

Overall path for decisions to come:

- Set time window and region filters.
- Read Price Level from Overview and Peak Hour from Train.
- Check in capacity, variable generation and interchange.
- Take action on tariffs, hedges or operations — as detailed in Scenarios C and B.

3.6.1 Scenario 2 — National retailer: NSW1 vs VIC1 in 2025

A nationwide retailer pondering 2026 tariffs and hedges is wondering if NSW and Victoria can have the same wholesale assumptions. The year and region filters are set to NSW1 and 2025 on the strategic dashboard; and views are updated without the need to extract.

For NSW1 in 2025, mean RRP is $103.3/MWh, mean demand about 7,497 MW, and the peak price hour is 17:00 at about $288/MWh. The correlation between RRP and capacity is −0.23, while the semi-scheduled generation is approximately 24% of the demand.

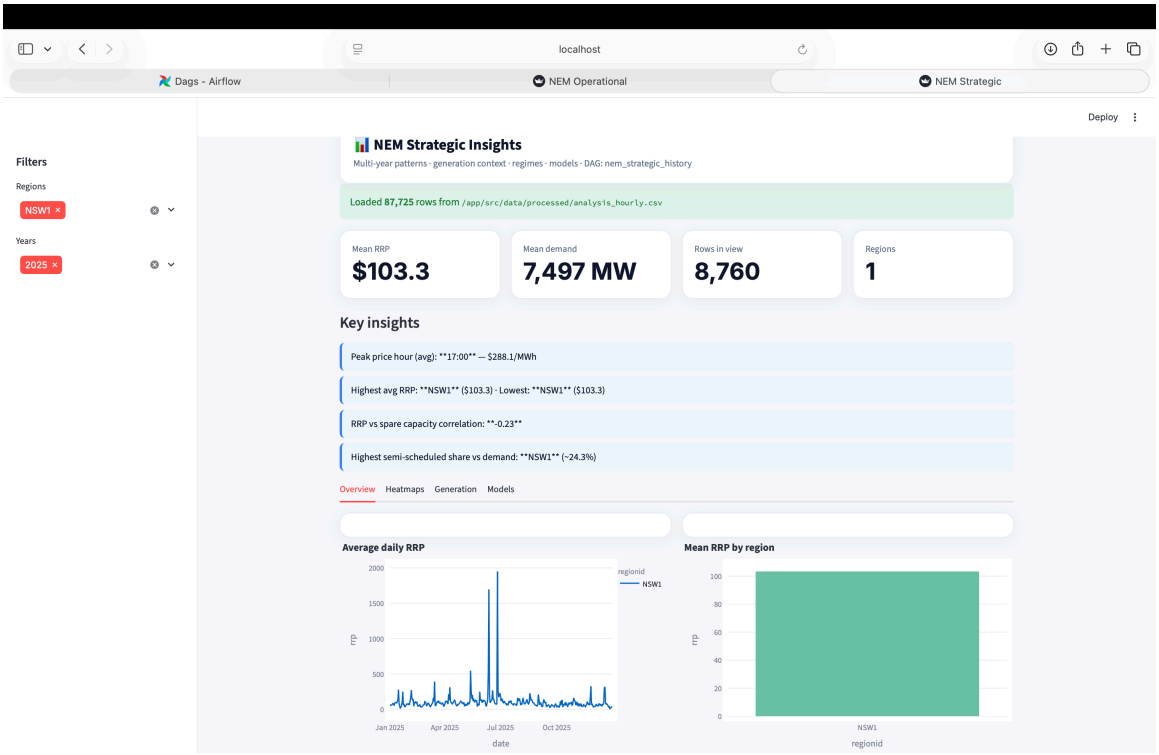


Figure 3.2. NSW1 · 2025 — overview KPIs and key insights.

The NSW heatmap reveals hot spots in the cost – the darker the color the closer to the average the cost is, with the most intense cells being in the middle of the week in the evening. Overnight/late morning blocks will be lighter, a direct input to peak/off-peak product design.

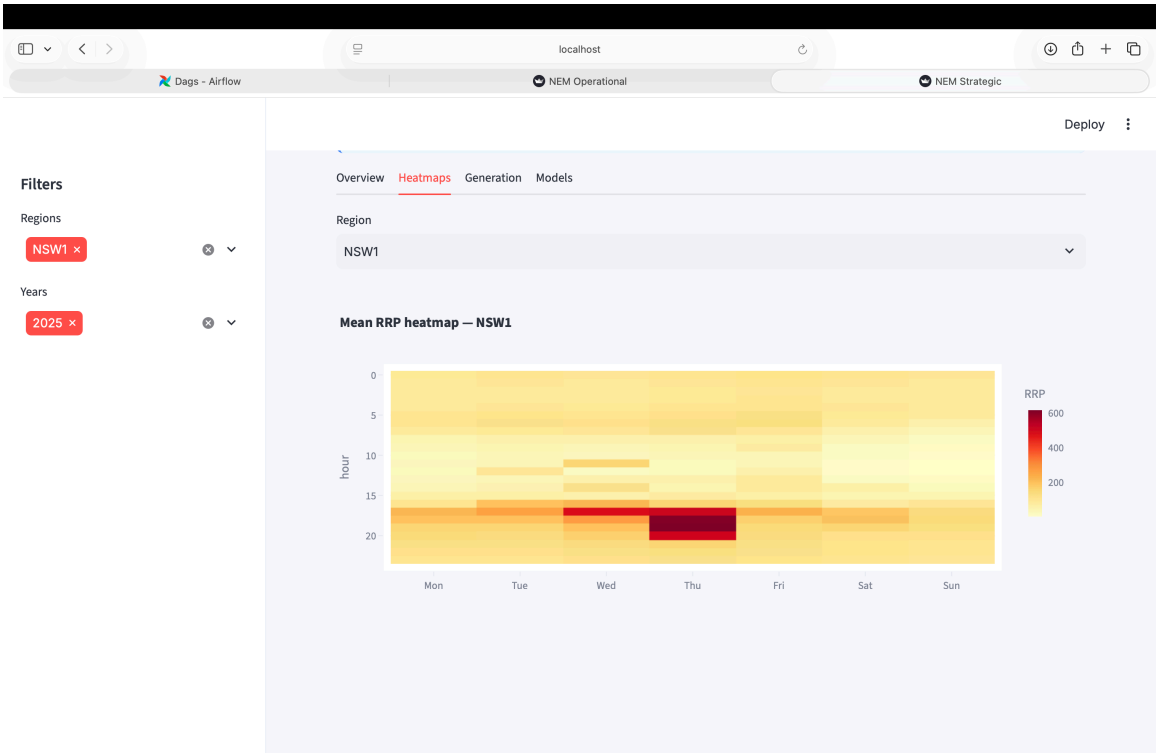


Figure 3.3. Mean RRP heatmap — NSW1 · 2025.

The same filters, the generation view reveals that there is a significant amount of dispatchable generation (around 7166MW in the highlighted bar), there is spare capacity, a distinct semi-scheduled generation contribution and a small negative mean net interchange bar (export-leaning in the aggregate).

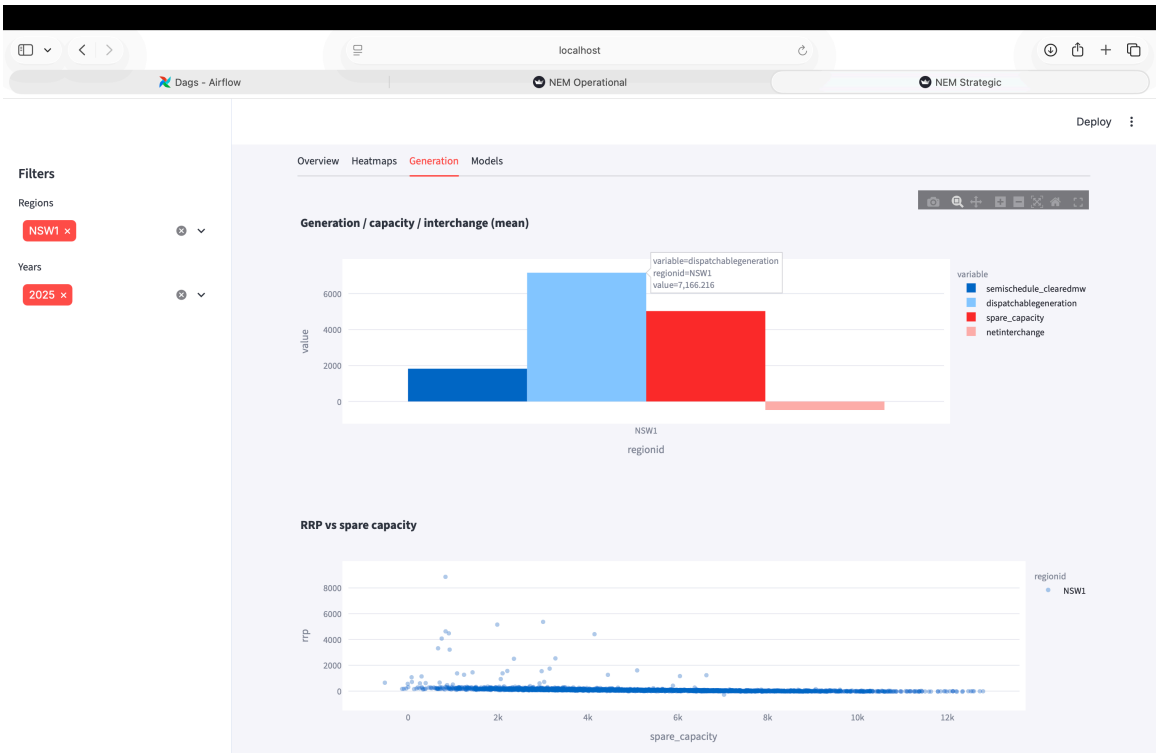


Figure 3.4. NSW1 · 2025 — generation, capacity, and interchange.

A change to commercial picture from the region filters only to VIC1 (year remains 2025) alters the commercial picture in an instant. The mean RRP drops to $77.9/MWh and the mean demand to approximately 4,881 MW for Victoria. The peak hour

is delayed (19:00), and at a lower peak average (~$203/MWh). The share of demand is higher than NSW (about 33%) and the RRP–spare capacity correlation is negative (≈ −0.29).

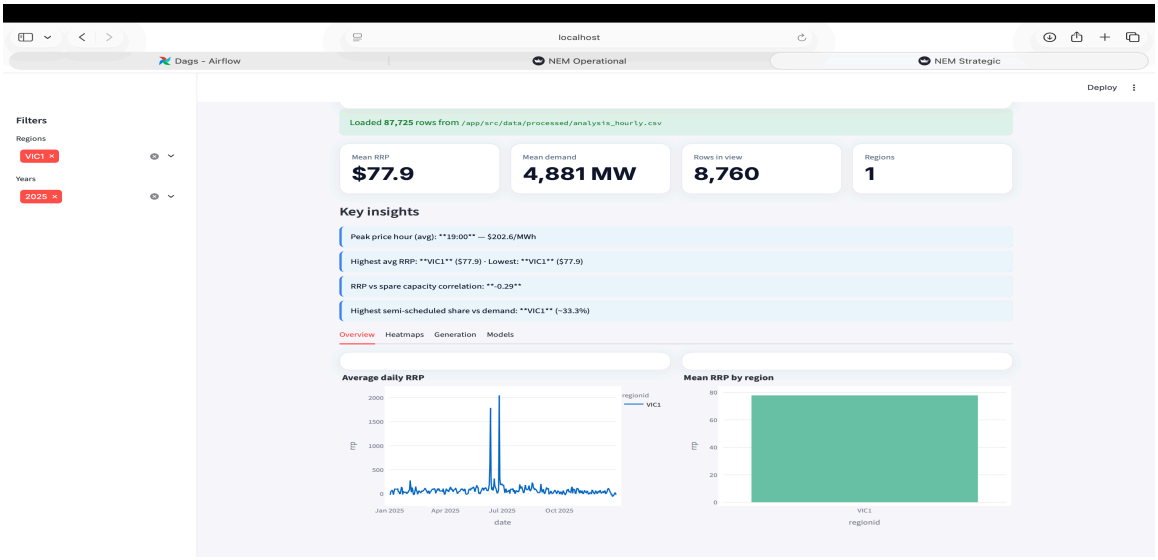


Figure 3.5. VIC1 · 2025 — overview KPIs and key insights.

Despite the high cost of the evening, the heatmap still shows it as being expensive in Victoria, but the level is not as high as NSW, which would reflect the lower average RRP.

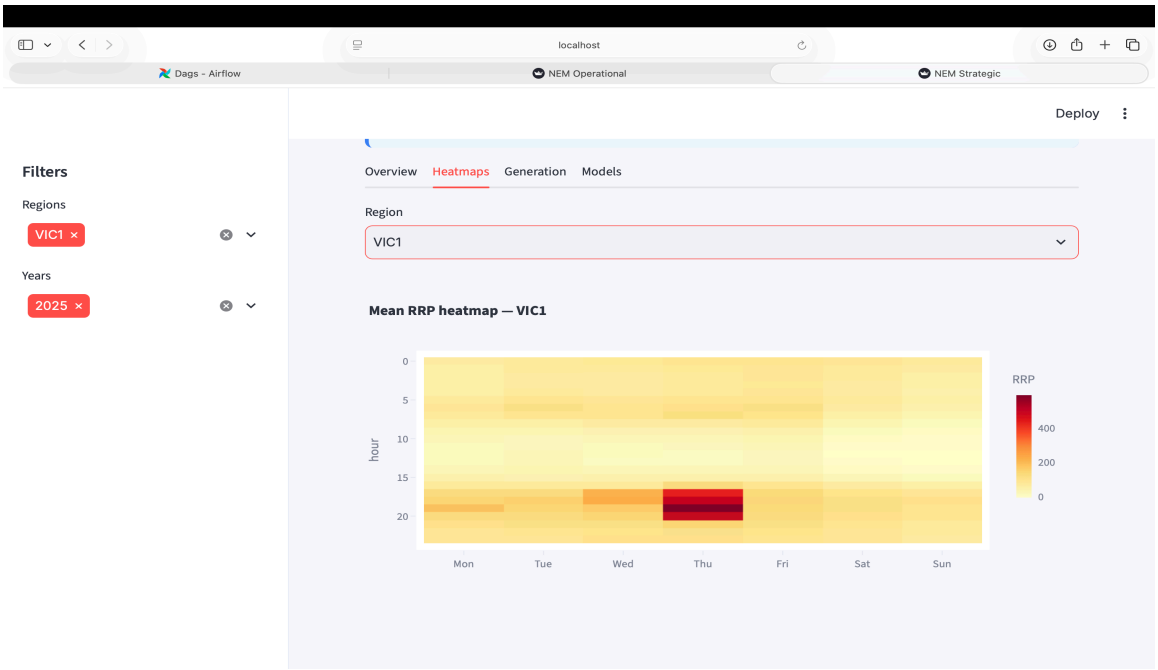


Figure 3.6. Mean RRP heatmap — VIC1 · 2025.

The results of the comparison that were filtered:

- Single national wholesale assumption under-prices NSW risk - NSW1 mean RRP ($103.3) is significantly higher than VIC1 ($77.9).
- NSW peaks earlier and harder (17:00, ~$288/MWh) than VIC (19:00, ~$203/MWh).
- Evening-aware products are needed for both states, with NSW requiring an increase in the peak premium and a greater level of hedge cover.
- Negative capacity correlations in both areas imply that risk desks should pay attention to spare capacity, rather than average RRP.

- VIC's share is higher than the semi-scheduled (33% vs 24%) which gives it a more VRE shaped risk but not the highest average price.
- Relevant decisions for the retailer:
- Higher peak component in NSW than VIC 2026.Tariffs are higher than VIC 2026.
- Hedges - heavier evening peak cover in NSW lighter but evening in VIC.
- Customer messaging — the early evening window of the day is the peak time in NSW, the overnight time is the cheaper shift period.

3.6.2 Scenario 3 — Operational desk: 1–2 January 2026

Peaking and firming controller asks after daily incremental load - which regions are important in the current window, what is the price doing, should the peekers be pre-started or held?

The operational dashboard for 1-2 January 2026 has a rather average average RRP of $36.2/MWh, and average demand is approximately 3,643 MW, but when you look at the maximum RRP in the same view, it is $230. Demand share is led by NSW (34.5%) and Queensland (33.7%), with Victoria at 20.6%. The mean net interchange in KPI strip is small (≈ 12 MW).

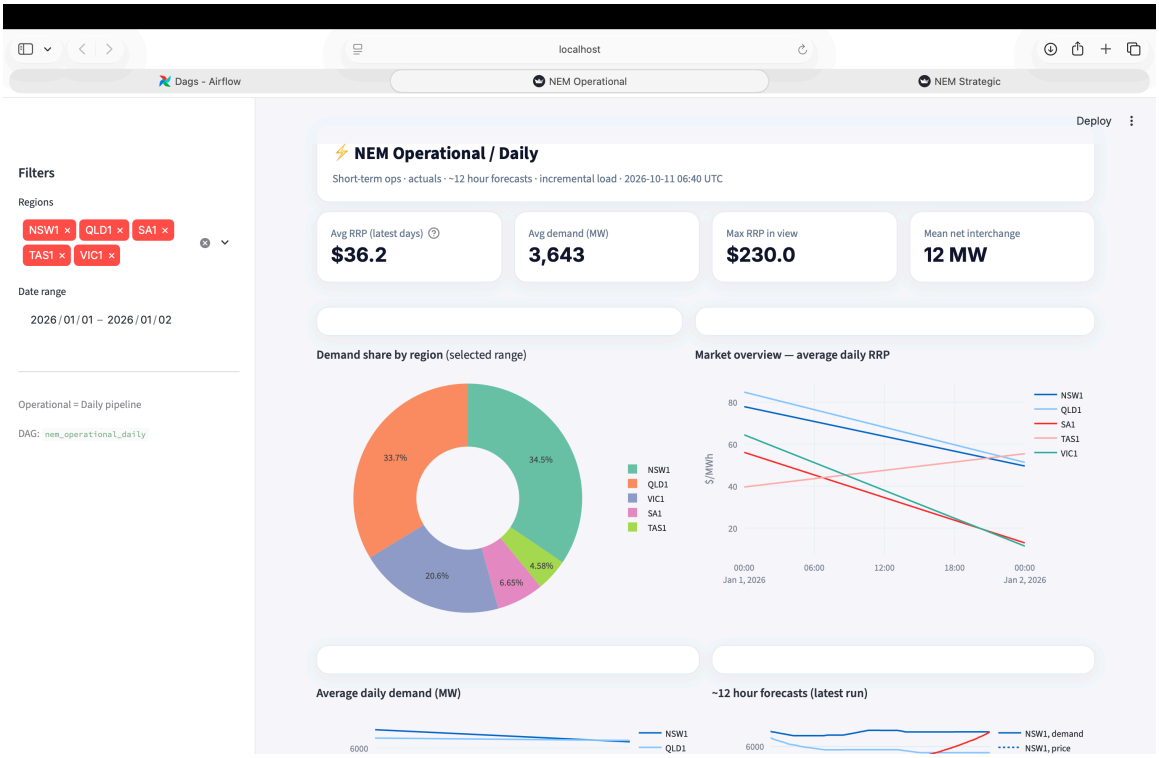


Figure 3.7. Operational / Daily dashboard — 1–2 January 2026.

The short horizon forecast panel indicates a divergence in the paths of regional demand, with an increase in Victoria demand, with several price series still relatively soft in the range plotted.

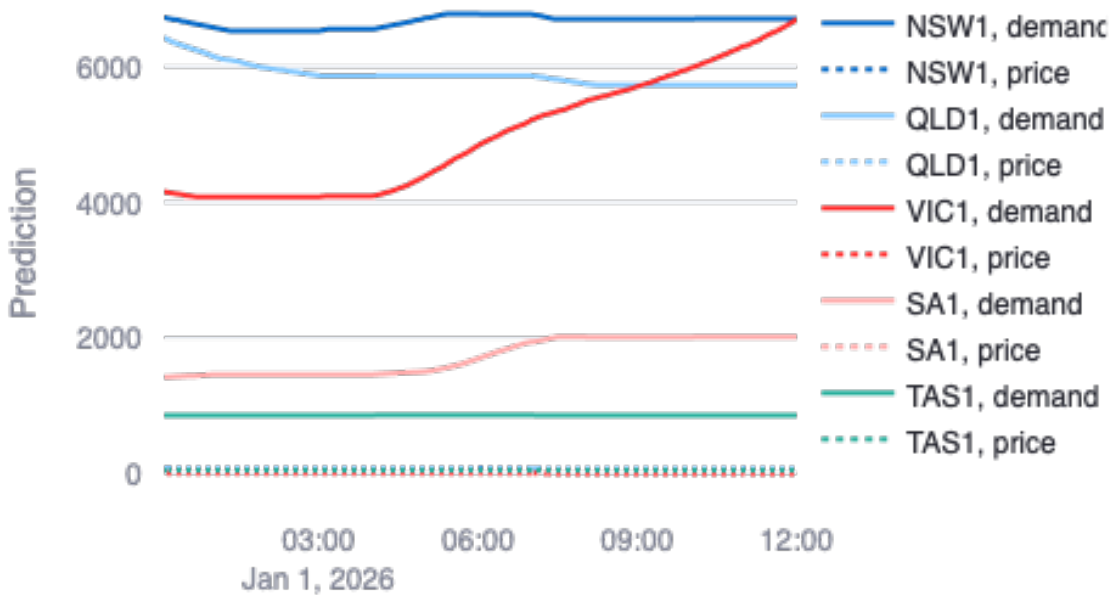


Figure 3.8. Approximate twelve-hour forecast panel from the latest operational run.

This is a window that can be operated:

- While the average price is not high, it is not negligible either, as can be seen from the maximum price of $230, which indicates that the risk of spikes is not gone.
- NSW and QLD are two-thirds of demand share – target these regions first.
- Broad peaker pre-start while forecast and average RRP stay soft.
- Maintain the readiness for NSW/QLD for fast start in case the price approaches the strategic heat map evening peak pattern.

3.7 Governance, ethics, privacy and security

Governance is the same as in previous team activities: weekly team coordination, version control through git and a tracking system called Jira for any remaining technical questions, with a project lead to take decisions on technical matters that are not resolved. Individual accountability regarding extract, warehouse, modelling and dashboard work, with feature branches and documented merge history. Contributor activity, the repository structure, the Jira board, and Teams coordination artefacts are provided in Appendix.

All inputs to the analytical method are publicly available AEMO NEMWEB market data through NEMOSIS. The pipeline doesn't handle any metering data for personal domestic use, customer identities or bids that are commercially confidential, except to the extent they are already available in market reports. That decreases the risk of privacy, but it does not take away the ethical obligation. Forecasts and dashboards are provided as decision support and not as automatic dispatch/automatic trading instructions. The restrictions regarding the behavior of the spikes, short time horizon, and local scheduling are mentioned to prevent the over-reading of model certainty.

Security is appropriate for a coursework deployment. The PostgreSQL credentials used in Docker Compose files are sample credentials and should not be used in any system exposed to the internet. The repository and runtime are designed for students in a local or controlled environment. For production deployment, it is necessary to implement secret management, access to the database is limited to the operator and analyst, and the retention time of the forecast logs needs to be outlined. Citing AEMO and library sources and having evaluation claims that correlate with stored metrics (on AEMO and the library), not only screenshots alone, are used to ensure academic integrity.

3.8 Limitations

There are some limitations in the data, choice of modelling, and deployment of the model.

Wholesale price spikes lead the way over error first. While it is sometimes helpful to use MAE or improvement over seasonal naive, point-forecast RRP is still quite limited. It's not a fair comparison to call the system a "precise" price oracle.

Second, there is a short window for forecasting. Operational outputs provide near-term readiness (hours) but not the multi-day unit commitment or pre-dispatch replacement. Strategic dashboards provide a history of structure, and do not estimate policy counterfactuals.

Third, orchestration is LOCAL. Each Airflow DAG and Streamlit app is deployed in a student machine using Docker Desktop.Airflow DAGs and Streamlit applications are deployed on a student machine using Docker Desktop. Schedules will only run when that host is running. This is okay for demo and marking, but not for 24/7 production.

Fourthly, generation and capacity views are based on market-reported fields made available at the extract (where applicable) such as non-uniform interchange and semi-scheduled context. Not a full network model – no numerical weather prediction, plant outage registers, nor bid stack reconstruction.

Fifth, training is still limited to events that could be reliably supplied and designed for the semester. Early lag windows must be dropped prior to linear models training and very recent AEMO intervals may not be available for public use.

The limits are given to evaluate the dual-dashboard product as a reproducible analytical system, and not as a ready-to-use platform for the market.

3.9 Future improvements

Efforts for future should be on evaluation, hosting, and feature richness in that order.

- Hosting — Run the operational DAG and database on an always-on host or a managed service, to avoid having to rely on the laptop being connected to do loads or refresh the Dashboard every day.
- Evaluation protocol — Publish a fixed walk-forward design in the repository, including the dates of the fold, the metrics, and also compared to the seasonal naive.
- Distinguish between short time horizons (e.g. 1 hour vs 12 hours) for MAE/RMSE reporting.
- Richer exogenous inputs — If allowed by licensing and effort, include absolute operational demand forecasts, weather and/or interconnector constraint indicators to check if price-spike error decreases.
- Optional deeper models — Measure before revisiting sequence models if they are deep — don't outrun the
- Marker demo path — Store a path that contains only the commands necessary to run the Docker path and has cached extract options for the core tables and dashboards so that assessors can repeat the same core tables and dashboards without downloading all the data for the full multi-year time period..

3.10 Proof of impact and professional demonstration

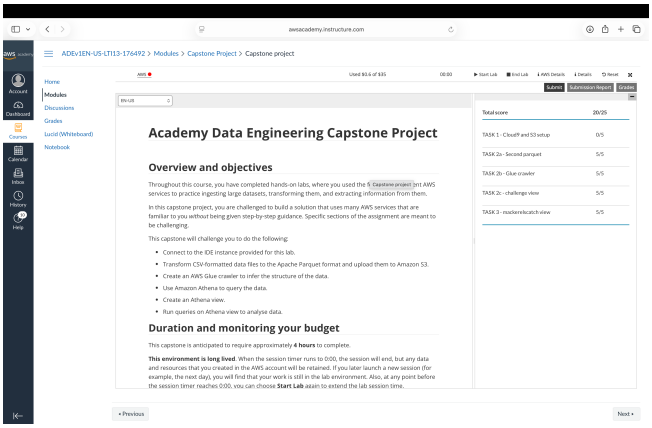
This project has an impact because a working product path is established, not because of uptake by external industry. The team delivered:

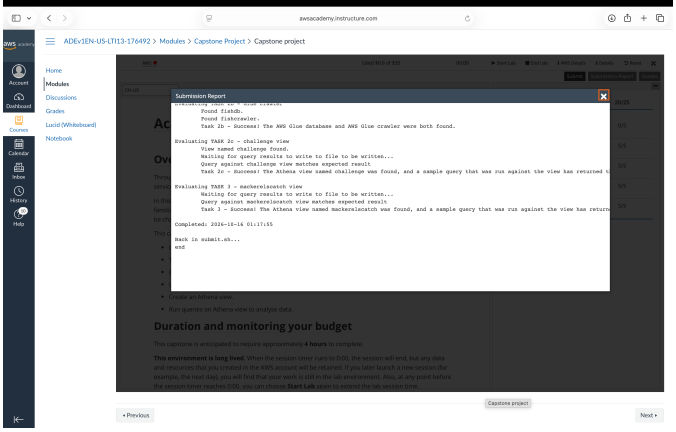
- Two Streamlit applications (one with and one without Airflow) along with a reproducible Docker Compose stack that includes PostgreSQL, as well as two Streamlit apps.
- Two pipelines (strategic multi-year history and operational, daily upsert) in real operating patterns.
- The opportunity to use interactive decision scenarios with live filters and measured KPIs such as NSW vs VIC in 2025, and 1–2 January 2026 operational window.
- An Assessment 3 demonstration video with the end-to-end narrative (https://youtu.be/DIrxi1R78ys).
- Multi-author GitHub and Jira history as professional "collaboration evidence".
- Wrote the architecture and core data model to make it easy for markers to understand the relationships between staging, features, forecasts and metrics.

The result is a relatively narrow but specific impact claim: the same warehouse can now respond to an impact over several years that can be compared to other warehouses, as well as to a question of operation start or hold on a same week basis without special spreadsheets. This is the improvement over Assessment 2 where there was technical loading but not much end-user decision support. There is a huge potential for wider organisational impact with hosted operations, stakeholder pilots and extended evaluation periods, all of which are future work and not claimed to be achieved.

Section 4 – Individual Work and Contribution Summary— Nitesh Raj Chaudhary Student ID: S394811

1. Individual Capstone Project completion evidence





2. Individual project contribution since Assessment 2

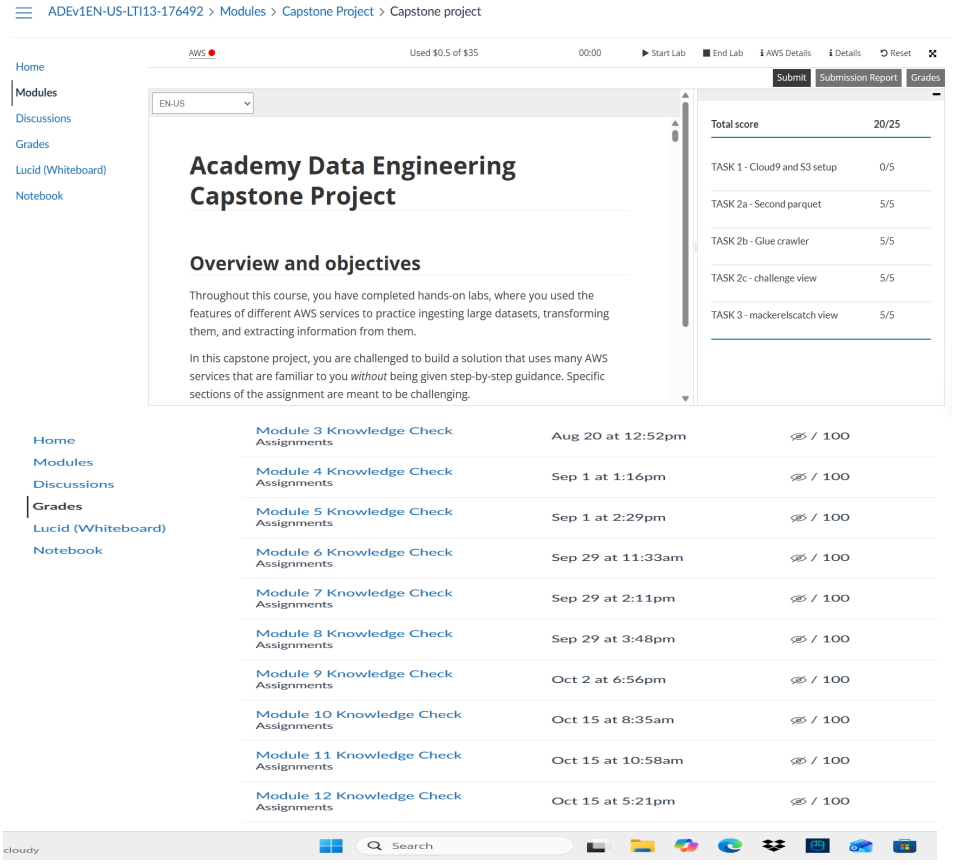
- Technical lead for Docker, PostgreSQL, NEMOSIS extract/clean integration, dual ETL scripts, Airflow DAGs, and README/SETUP documentation.
- The fields for extended price, demand, interchange and generation data were added to NEMOSIS extract/clean.
- Operational daily incremental pipeline scripts aligned with the ops + analytics logic (strategic history).
- Strategic history and daily load (daily analytics load wiring) are supported by Airflow DAGs.
- Daily incremental load and handling dates when extracting.
- Production stabilisation: host port 5433, postgresql+psycopg URL, daily upsert uniqueness, volume mounts for dashboard data.
- Repository maintenance and integration of strategic history and operational daily pipelines used by the whole team.
- Evidence: https://github.com/niteshS394811/S226-PRT661-DATA-SCIENCE-PRACTICE/commits?author=niteshS394811
- Jira project board: https://shrestharoshan931-1785816864381.atlassian.net/jira/software/projects/KAN/list

3. Brief professional reflection

The AWS labs introduced me to the concept of analytical databases, which store and compute separately, and demonstrated the need to put raw data in a staging area before curated models. Schema design and load practice were practical, not just theoretical because of the pattern-driven approach to working with Redshift. I also used that in our NEM project, creating a PostgreSQL layer for the cleaned AEMO extracts, another for the panel and features, and another for the forecast and metrics as datamart as opposed to a single table. I learned also from the group work that pipelines get screwed up on every little detail, such as a Docker port, driver URL, and daily upserts with windows that need to be aligned on midnight. Both Ridge and HistGradientBoosting were not as important as solving those problems. The labs and the project together have shaped my thinking on data systems beyond any doubt at the end of the model.

Individual Work and Contribution Summary — Dipesh Sedhai Student ID: S395457

1. Individual Capstone Project completion evidence



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2. Individual project contribution since Assessment 2

- Developed and documented the cleaning and feature-engineering path used for modelling (lags, rolling statistics, interchange, calendar fields).
- Model justification: seasonal naive, Ridge, and HistGradientBoosting, with explicit comparison to ARIMA and LSTM for this multi-region panel setting.
- Operational dashboard improvements and narrative linking near-term KPIs and forecasts to operator actions.
- Evidence: https://github.com/niteshS394811/S226-PRT661-DATA-SCIENCE-PRACTICE/commits?author=DipeshCdu
- Jira project board: https://shrestharoshan931-1785816864381.atlassian.net/jira/software/projects/KAN/list

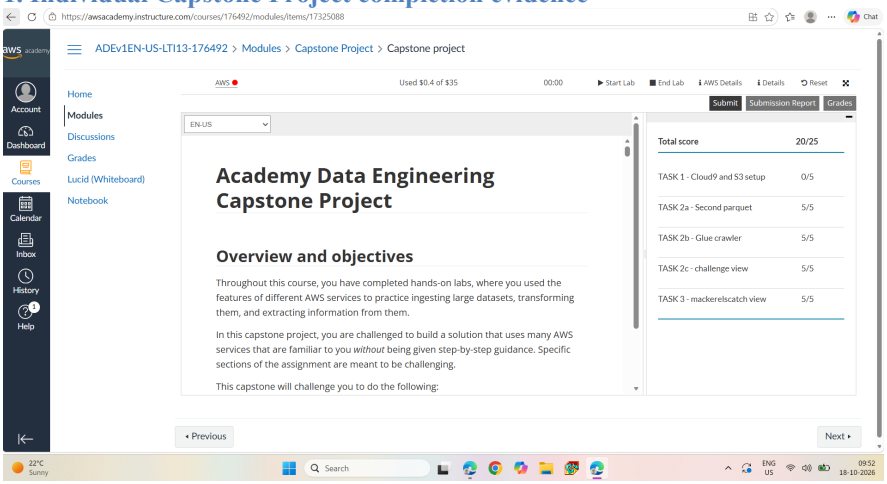
3. Brief professional reflection

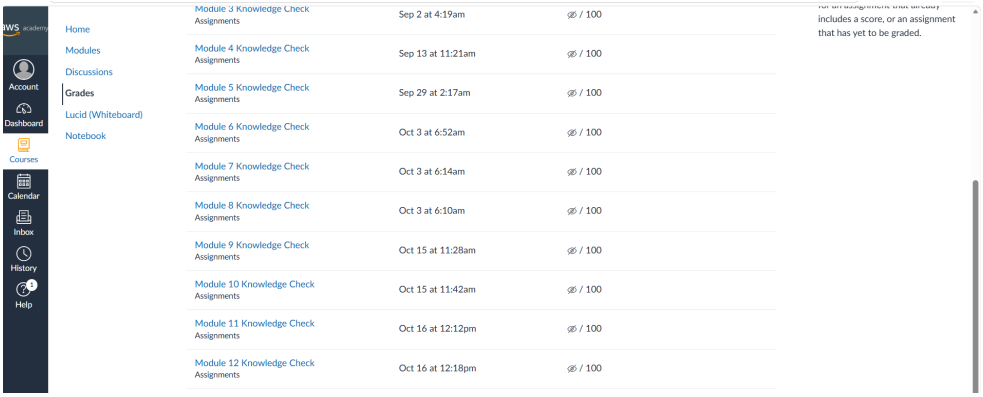
I concentrated on middle part of the pipeline - cleaning rules, lag features and why we have not tried to replace Ridge and HistGradientBoosting with LSTM. Lag_288 and rolling windows were not optional features as it became clear during the writing of the feature logic. I also had to rationalise with myself and others that the modest R^2 on price was good enough, provided the MAE was not worse than a naive baseline and provided there was some assistance from regimes for operators. Pushing the narrative of the action on the operational dashboard got me to connect charts to actions and not just report accuracy. The takeaway message was that feature design and proper error communication are as crucial as dressing up an algorithm in a cool hat.

Individual Work and Contribution Summary — Dilliram Dhakal

Student ID: S396633

1. Individual Capstone Project completion evidence





2. Individual project contribution since Assessment 2

- Owned the Strategic Insights dashboard (port 8502): region and year filters, KPI cards, and Overview, Heatmaps, Generation, and Models tabs.
- Implemented insight cards and action-oriented interpretation for energy operators, industry, and customers.
- Contributed to the strategic demonstration segment in the Assessment 3 group video.
- Evidence: https://github.com/niteshS394811/S226-PRT661-DATA-SCIENCE-PRACTICE/commits?author=DilliramDhakal
- Jira project board: https://shrestharoshan931-1785816864381.atlassian.net/jira/software/projects/KAN/list

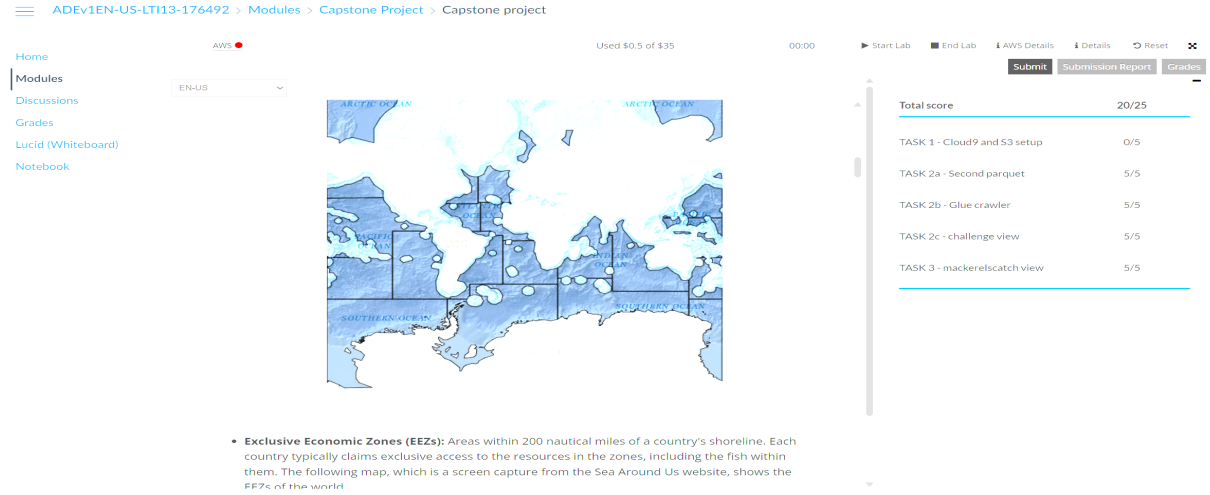
3. Brief professional reflection

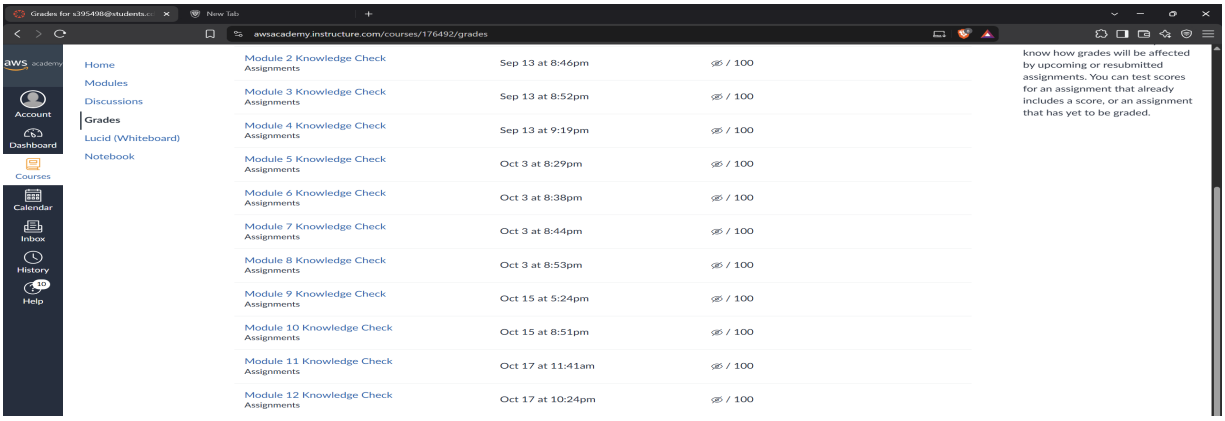
My focus was on the Strategic Insights app: filters, KPI cards, heatmaps and the generation views which display capacity and interchange. Early versions were like charts with no story, and feedback that the project had no end user insight led to the addition of insight cards and region-year filtering that updates the entire page. The NSW vs VIC comparison was a demo I had to learn about and realized that the value of a dynamic filter lies in the numbers changing in a manner a planner can respond to. I still find the dashboard layout difficult to navigate with too many tabs potentially obscuring the message, but associating each view with an operator, industry or customer action was very useful. If I had to do it again, I would create the decision questions before the charts so that the charts would be the object of the decision.

Individual Work and Contribution Summary — Roshan Kumar Shrestha

Student ID: S395498

1. Individual Capstone Project completion evidence





2. Individual project contribution since Assessment 2

- Strategic analytics: hourly rollups from five-minute data, price regime labels, and regression/classification metric tables.
- Loaded analysis frames into the PostgreSQL analytics schema for strategic queries and the Models tab.
- Supported group narrative on evaluation honesty under price spikes, limitations, and future work.
- Evidence: https://github.com/niteshS394811/S226-PRT661-DATA-SCIENCE-PRACTICE/commits?author=roshanshresthacdugit
- Jira project board: https://shrestharoshan931-1785816864381.atlassian.net/jira/software/projects/KAN/list

3. Brief professional reflection

Primarily worked on the analytics layer: rolling 5 mins to 1 hour, creating price regimes and creating metrics that the Models tab and the report could rely on in tables. When I forced Naive vs. Ridge vs. HistGB it was interesting to note that the R^2 was sacrificed during drastic price increases and yet the demand errors remained interpretable. Loading analytics data into the analytics schema gave me a glimpse of how simple it is for a dashboard to disagree with a notebook if they're not on the same table! I learned how to express limitations in an easily understood way to the group so that the group was not over-promising on the level of accuracy. I would like to have a fixed walk-forward script in the repo so that each retrain would have the same layout of the metrics.

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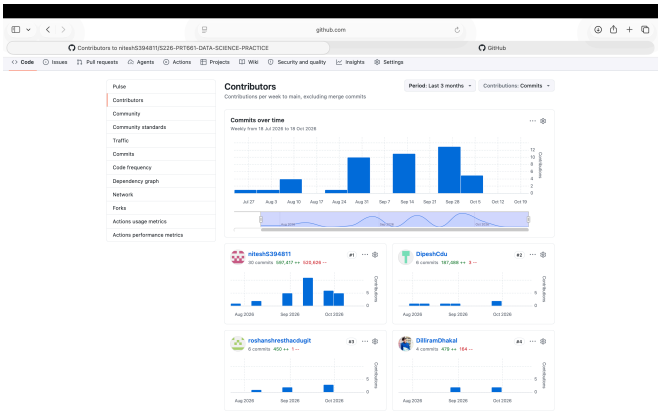
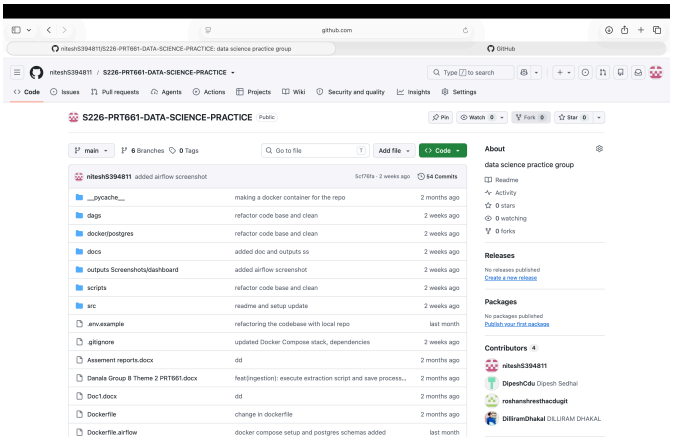
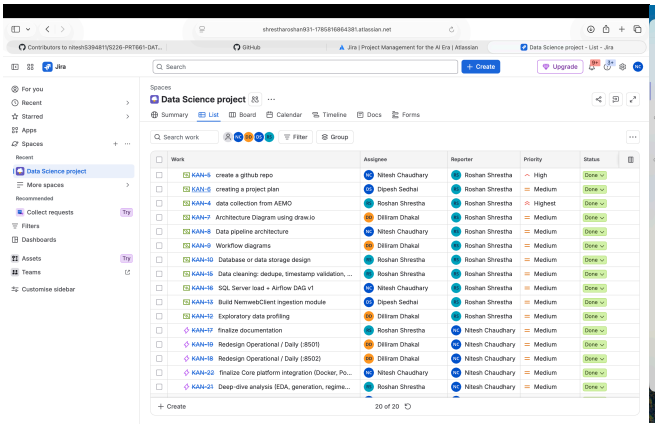
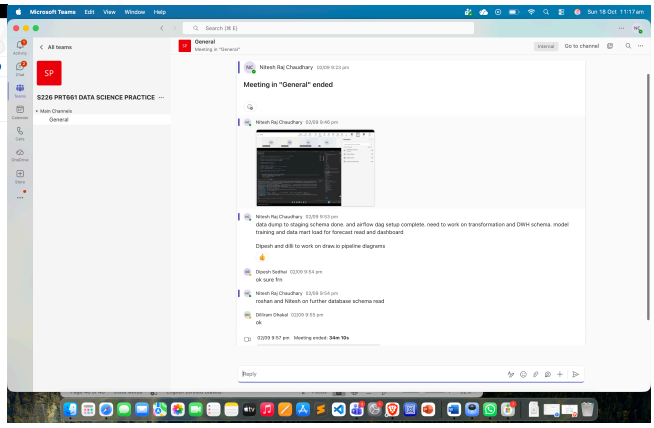
Appendix

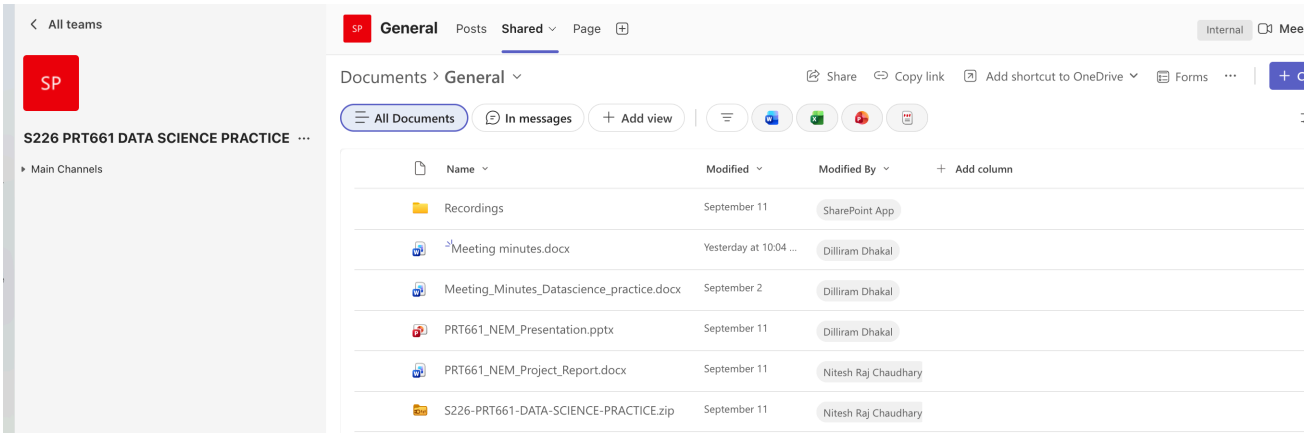
Contributor activity, the repository structure, the Jira board, and Teams coordination artefacts.



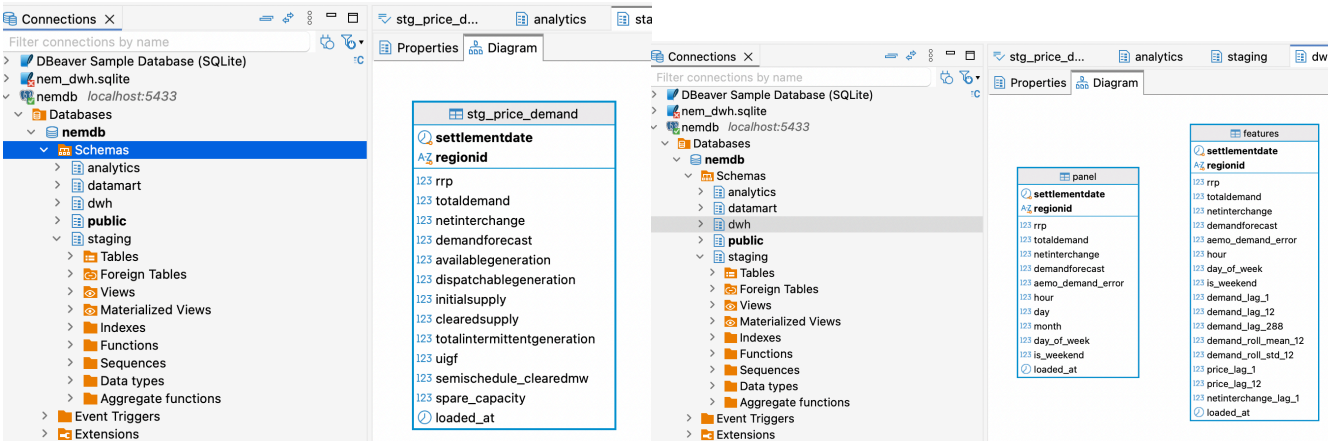


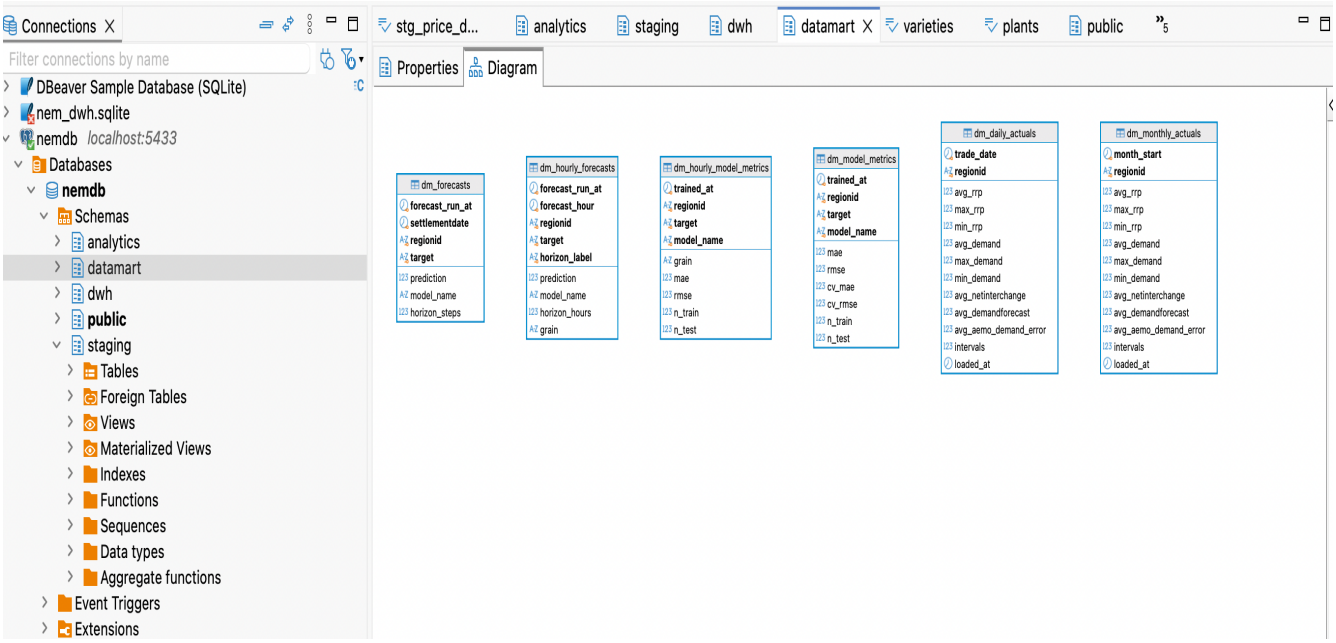


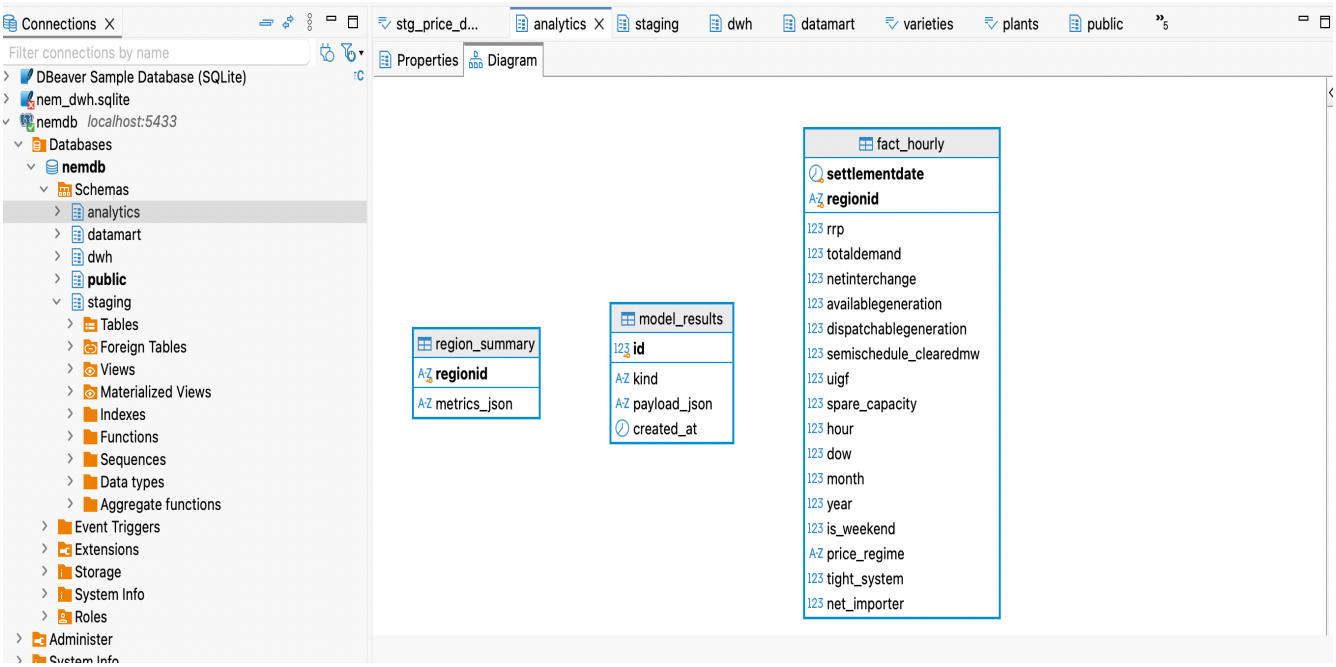




PostGres data schemas and apache airflow pipelines






Connections
Filter connections by name
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Databases
nemdb
Schemas
analytics
datamart
dwh
public
staging
Tables
Foreign Tables
Views
Materialized Views
Indexes
Functions
Sequences
Data types
Aggregate functions
Event Triggers
Extensions
Storage
System Info
Roles
Administer
stg_price_d...
analytics
staging
dwh
datamart
varieties
plants
public
Properties
Diagram
region_summary
regionid
metrics_json
model_results
id
kind
payload_json
created_at
fact_hourly
settlementdate
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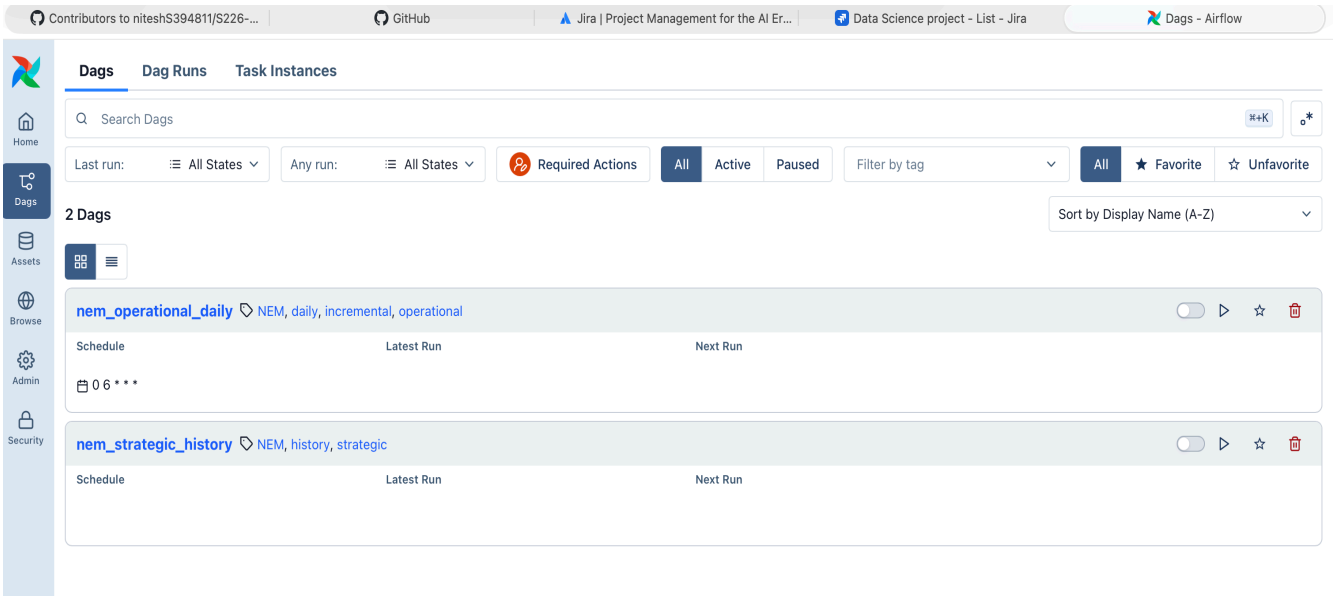
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Data Science project - List - Jira
Dags - Airflow
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